

The image shows a well-lit industrial networking lab. In the foreground, there are wooden-topped workbenches with black frames and multiple drawers. A black ergonomic chair is positioned in front of one of the benches. In the background, there are several racks filled with industrial networking equipment, including switches and routers, with various cables connected. A projector is mounted on the ceiling, and the room has a clean, professional appearance.

INDUSTRIAL NETWORKING

MENG 4010

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7B. EtherNet/IP Protocol

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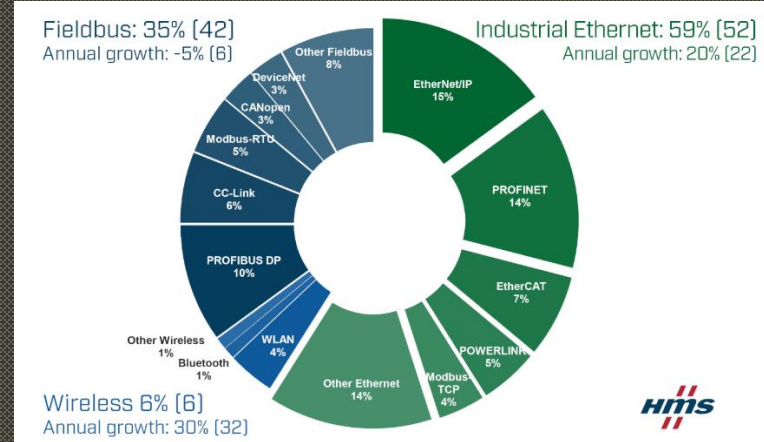
7B.1 Introduction to Industrial Ethernet and EtherNet/IP



<https://www.odva.org/technology-standards/key-technologies/ethernet-ip/>

- In the early days of industrial automation, control systems relied on serial communication protocols such as RS-232 and RS-485. These protocols were simple, low-cost, and reliable, but they were limited in speed, distance, and device count.
- To improve integration, industry developed fieldbus networks, which specialized communication systems for industrial devices. Examples include:

- Modbus RTU
- PROFIBUS
- DeviceNet
- CANopen



<https://www.fluke.com/en-ca/learn/blog/electrical/industrial-ethernet>

Speed is how fast data travels, while bandwidth is the maximum amount of data that can be sent at once. Like a highway, speed is how fast cars move, and bandwidth is how many lanes are available for traffic.

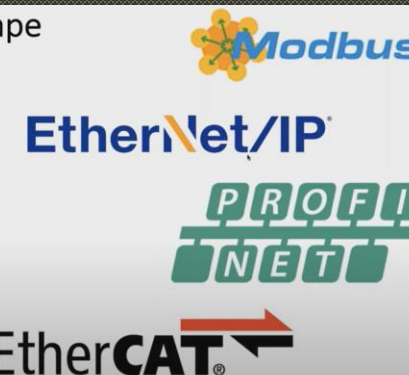
- While these fieldbus systems were effective, they often lacked interoperability, meaning devices from one vendor could not easily talk to devices from another.
- By the late 1990s, Ethernet technology, already dominant in office and IT networks, began to be adopted for industrial use. However, standard Ethernet (IEEE 802.3) was not designed for deterministic or time-critical control applications. Thus, Industrial Ethernet protocols emerged to bring real-time, reliable communication to factory environments.

7B.1 Introduction to Industrial Ethernet and EtherNet/IP

- EtherNet/IP (Ethernet Industrial Protocol) is an open, industrial network protocol that uses standard Ethernet and TCP/IP technology for communication between automation devices such as:
 - PLCs (Programmable Logic Controllers)
 - I/O modules
 - Drives and sensors
 - HMIs (Human-Machine Interfaces)
 - SCADA systems
- EtherNet/IP was developed by Rockwell Automation and managed by ODVA (Open DeviceNet Vendors Association). It is one of the most widely used industrial protocols worldwide.
- EtherNet/IP extends the Common Industrial Protocol (CIP), the same application-layer protocol used by DeviceNet and ControlNet, to run over standard Ethernet networks.

The Industrial Ethernet Landscape

- Since the late 1990s – Ethernet based networks have dominated the Industrial market
- Almost all Industrial Ethernet protocols are at least *partially* open
- Here are some of the top networks and their origin:
 - Modbus TCP (1990s) – from Modbus (1979)
 - Ethernet IP (2000) – from DeviceNet (1994)
 - Profinet (2003) – related to Profibus (1989)
 - EtherCAT (2007) – related to CANopen (1993)



The slide features four logos on the right side: Modbus (orange star logo), EtherNet/IP (blue text with an orange 'N'), PROFINET (green block letters), and EtherCAT (black text with a red arrow pointing right).

7B.1 Introduction to Industrial Ethernet and EtherNet/IP

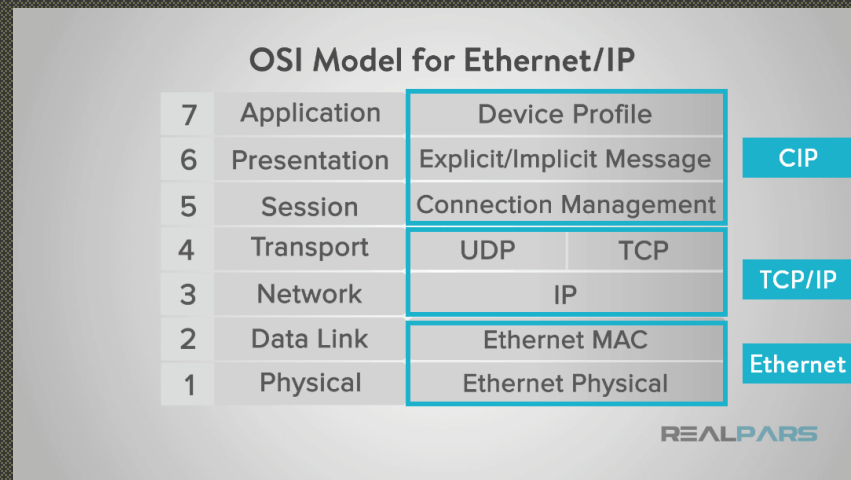
EtherNet/IP is essentially CIP over Ethernet. It defines:

- The object-oriented model for devices.
- Services for data access and control.
- The messaging types for communication.

EtherNet/IP basically specifies how those messages are transported using:

- TCP (Transmission Control Protocol) for explicit messaging.
- UDP (User Datagram Protocol) for implicit, real-time I/O messaging.

Analogy: Think of CIP as the language industrial devices use to communicate, while EtherNet/IP defines how the messages travel on Ethernet cables.



OSI Layer	Function in EtherNet/IP
Layer 1 (Physical)	Standard Ethernet cables, connectors
Layer 2 (Data Link)	Ethernet MAC addresses, frame handling
Layer 3 (Network)	IP addressing and routing
Layer 4 (Transport)	TCP/UDP communication
Layer 7 (Application)	Common Industrial Protocol (CIP)

EtherNet/IP operates across multiple layers of the OSI model, which means it can easily coexist with other TCP/IP-based applications

7B.1 Introduction to Industrial Ethernet and EtherNet/IP

A typical EtherNet/IP system includes:

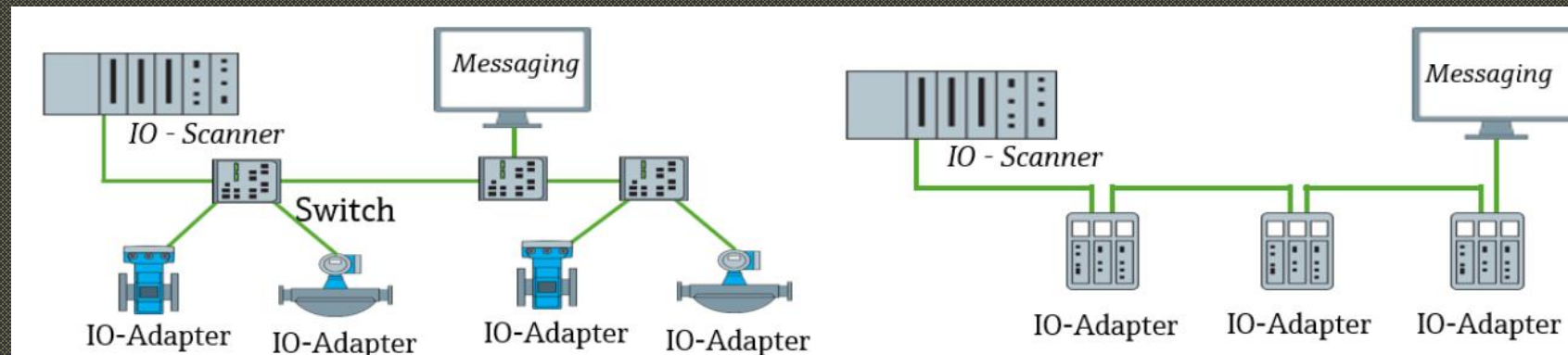
- Controller (e.g., Allen-Bradley ControlLogix PLC)
- I/O Devices (EtherNet/IP-enabled sensors, actuators, and drives)
- Network Switches
- HMI or SCADA
- Engineering Workstation (Studio 5000)

EtherNet/IP networks can be flat (single subnet) or segmented (multiple subnets connected via managed switches).

Managed switches are often preferred for diagnostics, VLANs, and traffic prioritization.

Feature	Benefit
Uses standard Ethernet hardware	Reduces cost and complexity
Open and vendor-independent	Interoperability between manufacturers
Scalable architecture	Suitable for small machines or large plants
Real-time I/O messaging	Supports time-sensitive control
Easy integration with IT systems	Enables data analytics, IIoT, and Industry 4.0 applications

EtherNet/IP networks can carry both control traffic (for machine operations) and information traffic (for data exchange with enterprise systems).



7B.1. Introduction to Industrial Ethernet and EtherNet/IP

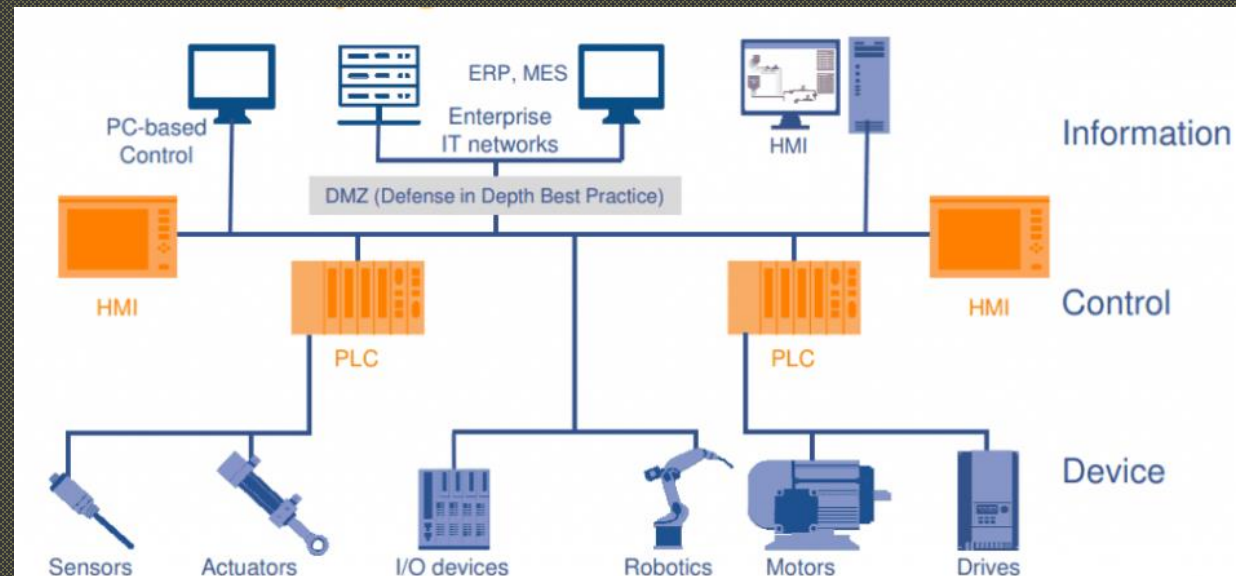
In a ControlLogix system, EtherNet/IP is typically used between:

- Controllers (ControlLogix or CompactLogix)
- Remote I/O modules connected through EtherNet/IP adapters
- Drives such as PowerFlex 525 with built-in EtherNet/IP ports
- HMIs such as PanelView Plus or FactoryTalk View SE

Using Studio 5000 Logix Designer, you can:

- Add a new EtherNet/IP module under the I/O Configuration tree.
- Specify the module's IP address.
- Configure the Requested Packet Interval (RPI) for I/O data updates.

EtherNet/IP devices communicate using standard RJ-45 connectors and Cat5e/Cat6 cables, but industrial environments often require ruggedized cables and connectors (M12 type).



7B.1. Introduction to Industrial Ethernet and EtherNet/IP

CIP uses two communication models to handle data:

- Client/Server Model → for explicit (non-real-time) communication
- Producer/Consumer Model → for implicit (real-time I/O) communication

1. Client and Server (Explicit Messaging)

- Client: the one who asks for something (initiates the request)
 - A client sends a message saying: “Please give me your input value,” or “Change this setting.”
- Server: the one who responds with the answer or performs an action
 - The server receives it, performs the request, and replies.

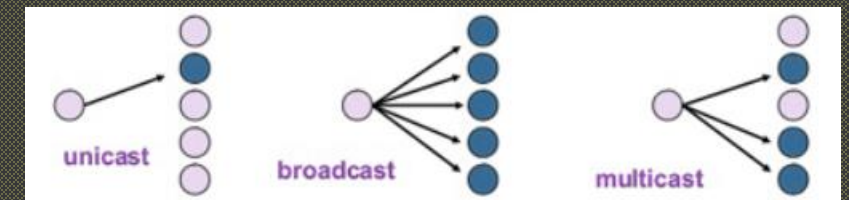
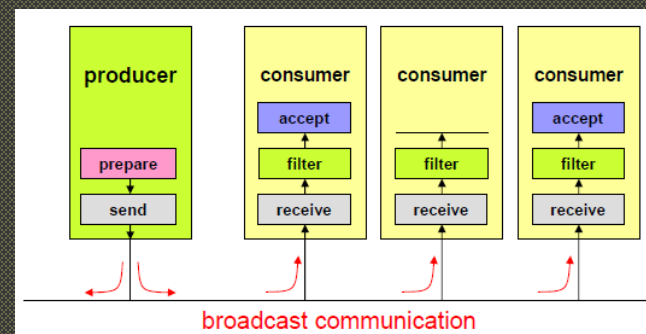
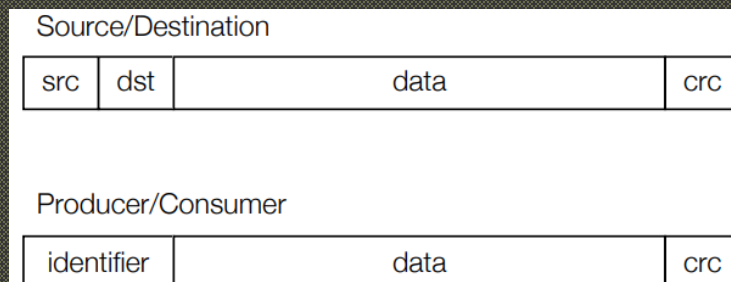
Example 1: PLC and Drive

- The PLC (Client) sends an explicit message like:
 - “VFD, please tell me your current speed parameter.”
- The VFD (Server) replies:
 - “The speed is 1450 RPM.”

This kind of message is not real-time – it is used for setup, configuration, or diagnostics.

Example 2: HMI and PLC

- The HMI (Client) requests a tag value:
 - “PLC, what is the tank level?”
- The PLC (Server) replies with the data.
- These are explicit messages, meaning they include:
 - The address of what is requested (e.g., tag or parameter)
 - The service type (read, write, reset)
 - And they wait for a response.



<https://www.esds.co.in/blog/difference-between-unicast-broadcast-and-multicast/>

7B.1. Introduction to Industrial Ethernet and EtherNet/IP

2. Producer and Consumer (Implicit Messaging)

- Producer: the one who sends out data (broadcasts it)
- Consumer: the one (or many) who receives and uses the data

This is for real-time I/O communication, it is fast and cyclic (happens continuously).

Instead of asking each time (like client/server), the producer just sends data repeatedly, and consumers listen if they are interested.

- CIP uses a Connection ID, not a “destination address.”

Each device decides:

“Is this data for me?”

If yes → it consumes it.

If no → it ignores it.

This is called multicast communication (one producer, many consumers).

Example 1: PLC and Multiple Drives

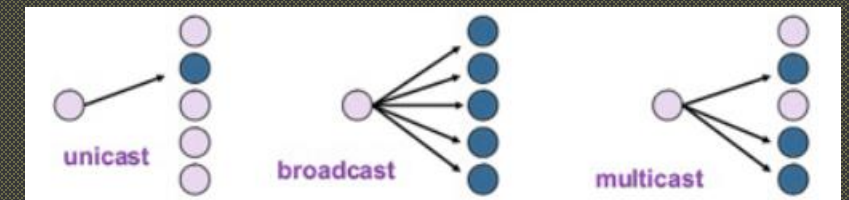
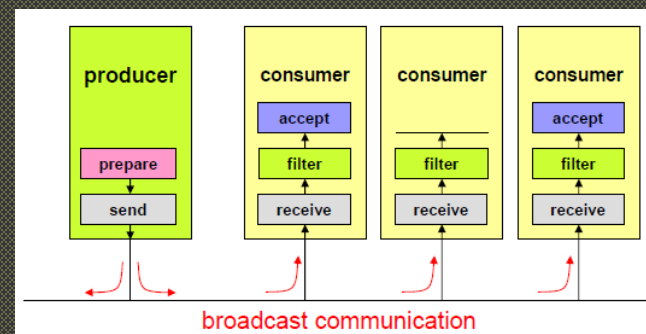
- The PLC produces output data (e.g., start/stop commands, speed reference).
- Several drives consume this data to run motors.
- Each drive knows, by the connection ID, that it should use the data.

Example 2: Sensor and PLC + HMI

- A temperature sensor (producer) sends temperature data every 100 ms.
- Both the PLC (to control heating) and the HMI (to display temperature) consume that data.
- The sensor is not interested who listens, it just keeps producing data.

That is the producer/consumer model in action.

Source/Destination			
src	dst	data	crc
Producer/Consumer			
identifier	data	crc	



<https://www.esds.co.in/blog/difference-between-unicast-broadcast-and-multicast/>

7B.1. Introduction to Industrial Ethernet and EtherNet/IP

- **Explicit Message:** Explicit messages contain addressing and service information that directs the receiving device to perform a certain service (action) on a specific part (e.g., an attribute of a given object) of a device.
- **Implicit (I/O) Message:** Implicit messages do not carry address and/or service information; any consuming nodes already know what to do with the data based on the connection ID that was assigned when the connection was established. Implicit messages are so named because the meaning of the data is “implied” by the connection ID. In most cases they are used to transport I/O data.

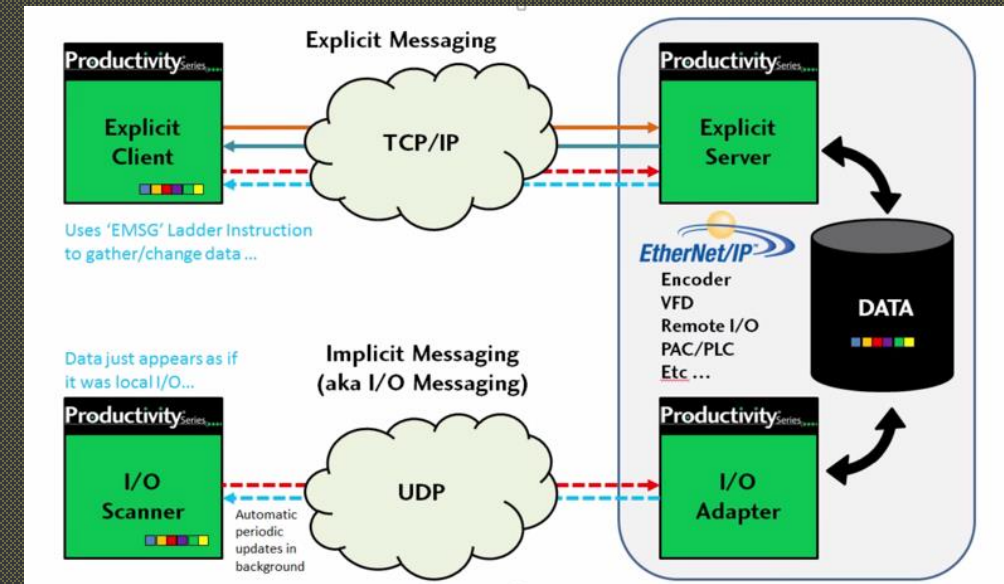
Feature	Explicit Message	Implicit (I/O) Message
Purpose	Configuration, diagnostics, setup	Real-time control data
Example	“Read parameter 45 from drive”	“Send motor speed data every 10 ms”
Communication Model	Client/Server	Producer/Consumer
Content	Includes addressing and service info	Meaning is implied by connection ID
Frequency	On-demand	Continuous (cyclic)
Example Devices	HMI, configuration tools	PLC, drives, sensors

1. When setting up a VFD:

- Explicit messaging is used to configure parameters (e.g., motor rated speed).

2. Once running:

- The PLC and VFD use implicit messaging to exchange control and status data every few milliseconds.



<https://library.automationdirect.com/ethernetip-implicit-vs-explicit-messaging/>

7B.1. Introduction to Industrial Ethernet and EtherNet/IP

Terms used with CIP

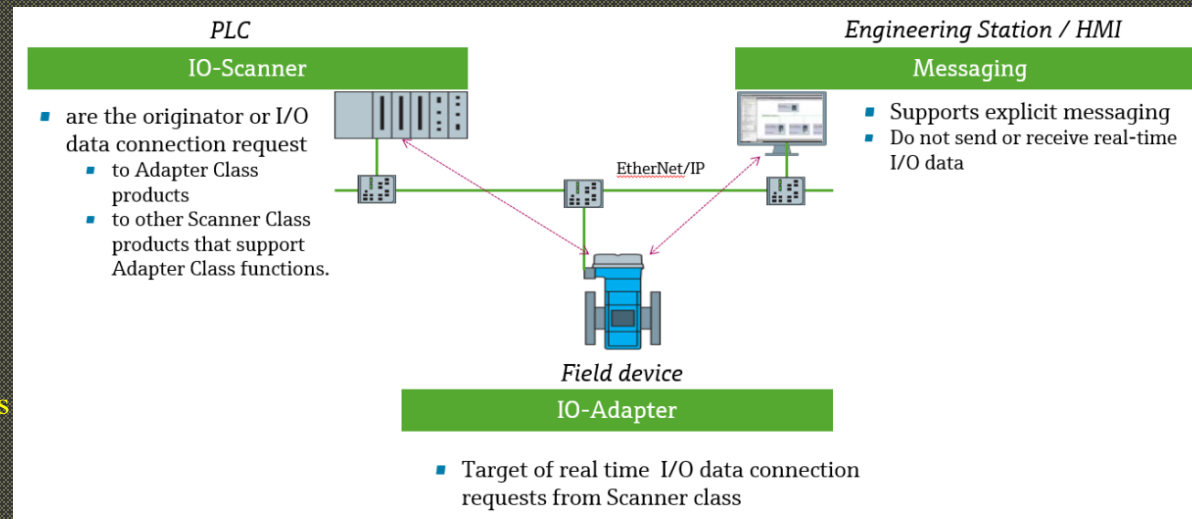
- **Scanner:** Cyclically maps real-time I/O data across the network. Examples: PLCs / controllers.
 - It opens connections and initiate message transfers. Can serve as both originators of output data and targets for devices that want to send them output data.
- **Messaging:** Handles explicit (non-real-time) communications. Examples: diagnostics, configuration, SCADA, HMI.
 - It can act both as server and client (initiate or respond to explicit messaging).
- **Adapter:** Provides device-specific functions using EtherNet / IP protocol. Examples: sensors, valves, gateways.
 - It supports both Explicit and Implicit operations.

Scanner

- The master device, usually the PLC.
- It initiates and maintains all I/O connections.
- It scans the network cyclically to exchange real-time I/O data.
- It can also send explicit messages (for setup or monitoring).

Example:

- CompactLogix PLC = Scanner
- It polls all connected devices (adapters like sensors, drives, etc.) regularly to update I/O tables.

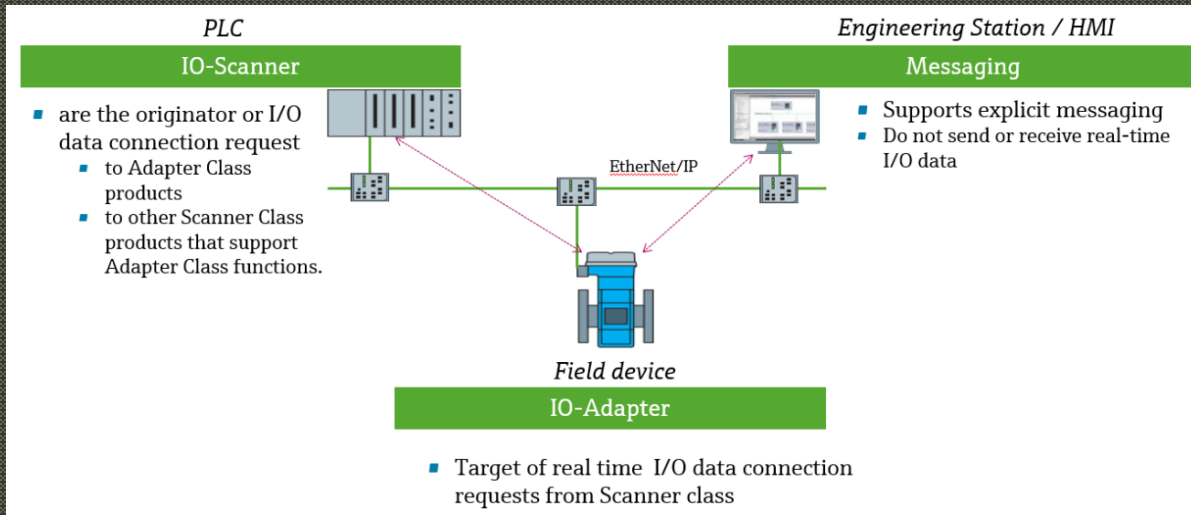


Adapter

- The field device: sensor, valve, drive, or I/O block.
 - Provides the actual input/output data to the network.
 - Supports both implicit (I/O) and explicit messaging.
- Example:
- PowerFlex Drive = Adapter
 - Sends motor status (input)
 - Receives speed commands (output)
 - Photoelectric sensor = Adapter
 - Sends ON/OFF status

7B.1. Introduction to Industrial Ethernet and EtherNet/IP

Role	Function	Common Devices	Messaging Type
Scanner	Initiates communication, controls I/O	PLC, PAC	Implicit + Explicit
Adapter	Provides data, responds to PLC	Drive, Sensor, Valve	Implicit + Explicit



<https://netilion.endress.com/blog/ethernet-ip/>

Device	Role	Description
PLC	Scanner / Client / Producer	Controls everything; sends outputs; requests setup data
Drives (VFDs)	Adapter / Server / Consumer	Receive speed commands, send feedback
Sensors	Adapter / Producer	Send input data (limit switch ON/OFF)
HMI	Client / Consumer	Reads PLC tags, displays values
SCADA System	Client / Consumer	Collects data from many PLCs for monitoring

Scenario	Who is Client/Server	Who is Producer/Consumer	Message Type	Description
HMI reads a PLC tag	HMI (Client) → PLC (Server)	—	Explicit	On-demand read of tank level
PLC configures drive parameter	PLC (Client) → Drive (Server)	—	Explicit	Setup of speed limits
PLC sends motor start command	PLC (Producer) → Drive (Consumer)	PLC→Drive	Implicit	Real-time control signal
Sensor sends temperature	Sensor (Producer) → PLC & HMI (Consumers)	Sensor→PLC,HMI	Implicit	Real-time data to both
PLC monitors drive fault	PLC (Consumer) ← Drive (Producer)	Drive→PLC	Implicit	Real-time feedback

7B.2 Fundamentals of Common Industrial Protocol (CIP)

The Common Industrial Protocol (CIP) is the core of EtherNet/IP.

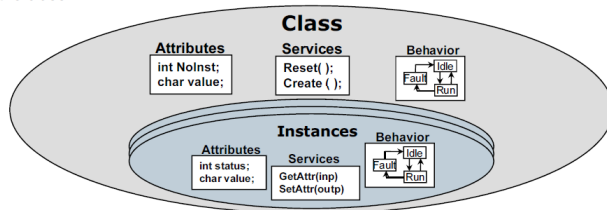
CIP is an object-oriented, connection-based, application-layer protocol that provides a standardized way for industrial devices to exchange information and control data.

CIP defines:

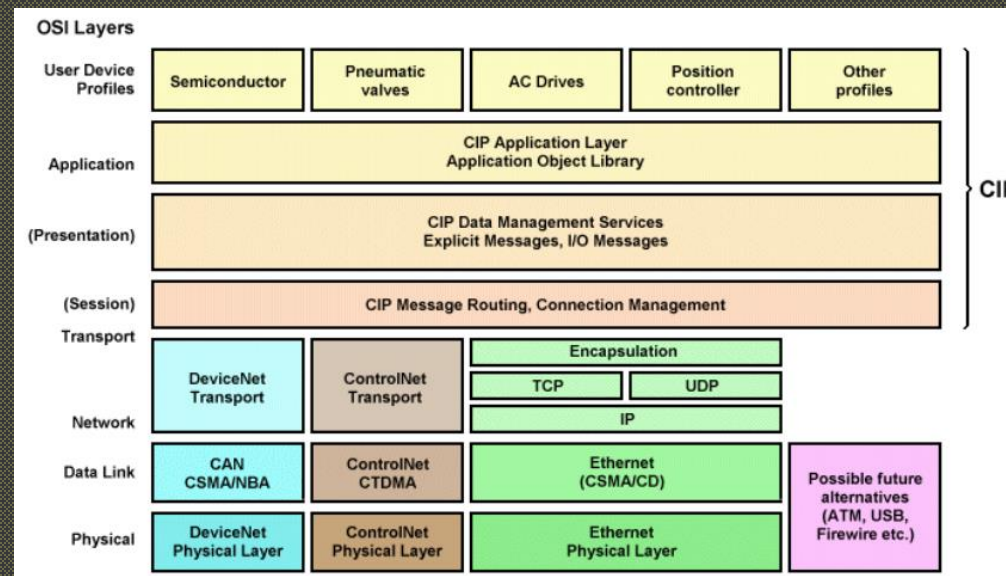
- The data model used by all devices.
- The services and messages exchanged between devices (actions that can be performed e.g., read, write, reset).
- The behavior of devices on the network (how the object responds to certain commands).

Since CIP is used not only in EtherNet/IP but also in DeviceNet, ControlNet, and CompoNet, it provides a common language across different physical networks.

- Each distinct type of object belongs to a specific *Class*
- Objects that belong to the same class are called *Instances* of that class
- Data items within an object are called *Attributes*
- All attributes can be addressed with Class:Instance:Attribute IDs
- *Services* are explicit tasks that an object performs
 - These may be directed at a specific instance or at the class, which affect all instances of the class



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7B.2 Fundamentals of Common Industrial Protocol (CIP)

Everything inside a device can be described and accessed as an object.

Think of a CIP object like a “folder” in a computer, which contains information (data) and actions (commands) about a specific function of a device.

Each object represents something meaningful in the device.

For example:

- An Identity Object tells you who the device is (vendor, model, serial number).
- An Analog Input Object gives you real-time sensor values.
- A Connection Object handles how data is sent between devices.

So, a device (like a motor drive or an I/O module) is just a collection of CIP objects.

Object Name	Class ID	Function	Example Attributes	Example Services
Identity Object	0x01	Device info	Vendor ID = 100, Product Name = “TempSense3000”	Get_Attribute_Single
Assembly Object	0x04	Input/Output data	Input = Temperature value, Output = Configuration bits	Set_Attribute_Single
Connection Object	0x05	How it communicates	RPI = 50 ms, Connection Type = Multicast	Open/Close Connection
TCP/IP Interface Object	0xF5	IP settings	IP = 192.168.1.50, Subnet = 255.255.255.0	Set_Network_Config

7B.3 CIP Object Model

Each object or component is accessed using a structured, uniform address format:

- Node Address: Unique per device (MAC ID for DeviceNet/ControlNet, IP for EtherNet/IP) (e.g., 192.168.1.10)
- Class ID: Identifies object class (e.g., 0x01) In CIP, classes define the type of data or functionality within a device.
- Instance ID: Identifies object within the class (e.g., 0x01)
- Attribute ID: Identifies specific attribute (e.g., 0x04)
- Service Code: Indicates the action to perform (e.g., 0x0E)

0x - prefix for a hexadecimal numeric constant.

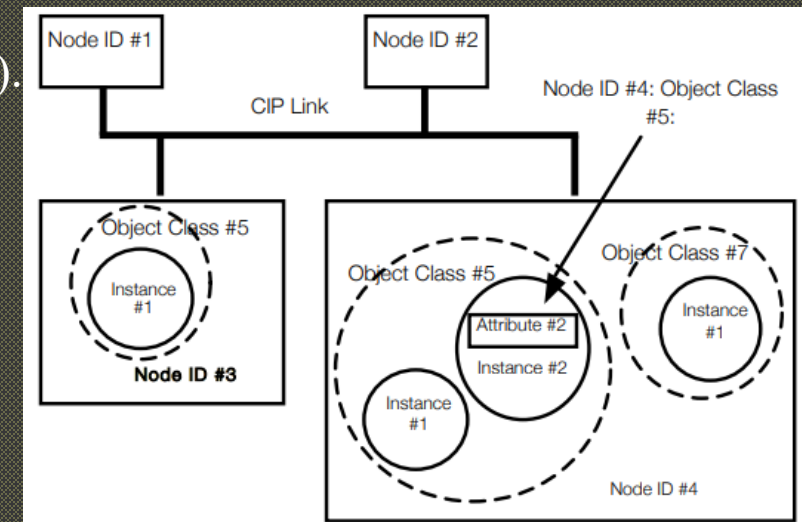
Object Types

- Public Objects: Standardized by CIP (e.g., 0x0000 - 0x0063, 0x00F0 - 0x02FF)
- Vendor-Specific Objects: Custom-defined within reserved ranges (e.g., 0x0064 - 0x00C7, 0x0300 - 0x04FF)

Address Widths

- Most use 8- or 16-bit addresses.
- 32-bit addresses reserved for future use (mainly instance addressing).

Only objects modeled in CIP are accessible over the network.



7B.3 CIP Object Model

Object = Attributes + Services + Connections

Each CIP object is made up of three key parts:

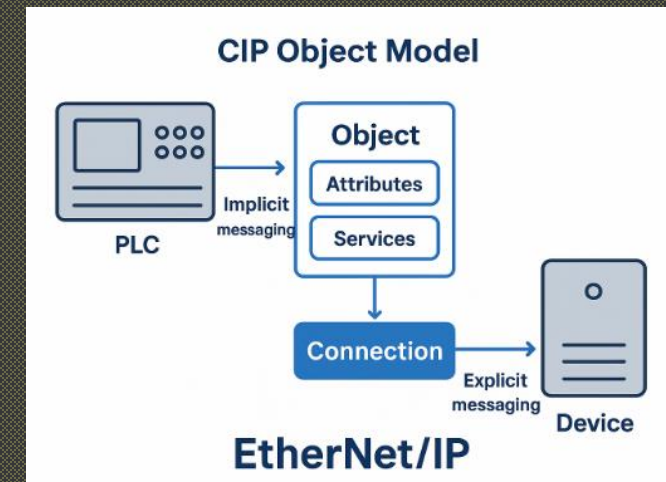
Part	What it is	Example
Attributes	Data values that describe the object	“Vendor ID = 100”, “Temperature = 25.6 °C”
Services	Actions (commands) you can perform	“Reset device”, “Read temperature”, “Change configuration”
Connections	Defines how and when data is exchanged	“Send data every 100 ms via implicit messaging”

1. Attributes: The Data of the Object

Attributes are just variables stored inside the object.

They describe the current state or configuration.

Attribute Name	Value	Meaning
Vendor ID	100	Company ID
Device Type	12	Analog Sensor
Product Code	3000	Model Number
Serial Number	12345	Unique ID
Product Name	TempSense3000	Name shown in software



7B.3 CIP Object Model

Object = Attributes + Services + Connections

2. Services: What You Can Do

Services are like commands you can send to the object.
Each service has a number (Service Code) and a function.

Service Code	Name	Description
0x01	Get_Attributes_All	Reads all attributes in the object
0x0E	Reset	Resets the device
0x10	Set_Attribute_Single	Writes to a specific attribute
0x0E	Get_Attribute_Single	Reads one attribute

3. Connections: How Data Is Sent

Connections describe how data flows between devices.
There are two kinds:

- Explicit Connection:
 - Used for reading/writing attributes occasionally.
 - Example: Read firmware version or change IP address.
- Implicit Connection:
 - Used for continuous, real-time I/O data.
 - Example: PLC receives new sensor readings every 50 ms automatically.

Each connection has parameters like:

- Direction (input/output)
- RPI (Requested Packet Interval)
- Transport type (multicast/unicast)

Analogy
Recap

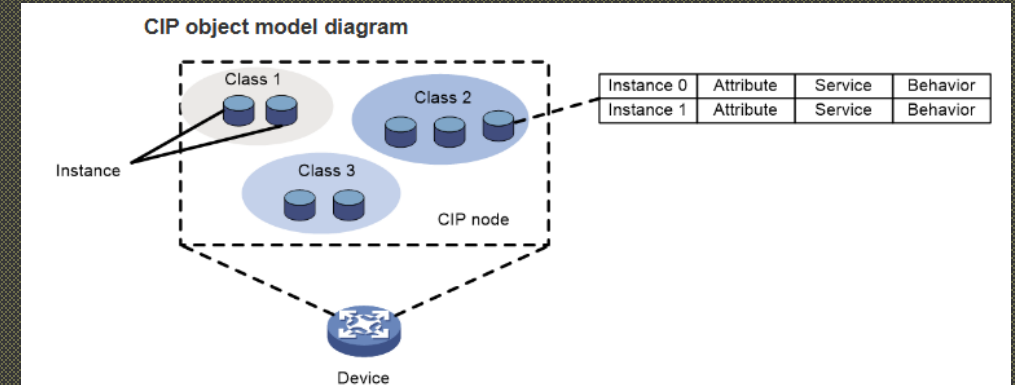
Real-Life Example	CIP Equivalent
Car dashboard	Device
Speedometer	Object
Current speed	Attribute
“Reset trip” button	Service
Bluetooth link to phone	Connection

7B.3 CIP Object Model

EtherNet/IP simply uses CIP objects over Ethernet. So, EtherNet/IP sends CIP messages inside Ethernet packets.

These packets contain:

- Object ID (which object we are talking to)
- Instance ID (which copy of the object)
- Attribute ID (which specific data point)
- Service Code (what we want to do)



https://www.h3c.com/en/d_202408/2239284_294551_0.htm

Example: A PLC wants to read the current temperature from a sensor at 192.168.1.50.

- PLC sends an Explicit Message (TCP):
 - Service: Get_Attribute_Single
 - Class ID: Assembly Object (0x04)
 - Instance: 1
 - Attribute: 3 (Temperature)
- Sensor responds:
 - Temperature = 25.6 °C

- **Class ID (Object ID):** A number that identifies a type or group of similar objects.
Example: The Identity Object (Class ID = 0x01) includes general information like the vendor ID and serial number.
- **Instance ID:** A number that identifies one specific example of that class.
Example: A multi-axis drive may have several Motor Objects, one for each moto, each with a different instance ID.
- **Attribute ID:** A number that identifies a specific piece of data inside an instance.
Example: In the Identity Object, one attribute could be the product name.
- **Service Code:** A number that tells what action to perform on the object.
Example: Get_Attribute_Single reads a value, and Set_Attribute_Single writes a value.

The PLC just used CIP over Ethernet (EtherNet/IP) to talk to the sensor.

7B.3 CIP Object Model

Example: ControlLogix communicating with an EtherNet/IP Motor Drive (PowerFlex 525 drive).

1. Inside the PowerFlex Drive (CIP Objects):

Object	Description	Example Attributes
Identity Object (0x01)	Basic device info	Vendor ID = 2 (Rockwell), Product Name = “PowerFlex525”
Assembly Object (0x04)	I/O data	Input = Drive status, Output = Speed command
Parameter Object (0x0F)	Drive parameters	Speed, Frequency, Fault code
Connection Object (0x05)	Defines I/O update rate	RPI = 10 ms

1. Identity Object

- When the PLC connects to the drive, it first reads the Identity Object (Class 0x01).
 - Attribute 1: Vendor ID → Rockwell Automation
 - Attribute 2: Product Type → AC Drive
 - Attribute 3: Revision → 3.02
- This helps the PLC recognize what kind of device it is communicating with.

2. When the drive is added in Studio 5000:

- The PLC automatically creates a CIP connection (implicit messaging).
- The PLC sends the speed command (output data).
- The drive sends back status and feedback (input data).
- You can also use explicit messages to change settings (like acceleration time or control mode).

3. Parameter Object

- The Parameter Object (Class 0x0F) holds adjustable settings.
 - Instance 1 = Max Speed
 - Instance 2 = Acceleration Time
 - Instance 3 = Control Mode
- PLC can use explicit messages to read/write these parameters.
 - Example: PLC writes Accel Time = 5 seconds.

2. Assembly Objects

- The PLC then looks for Assembly Objects (Class 0x04) that define I/O data.
 - Input Assembly → feedback data from drive (speed, current, status bits)
 - Output Assembly → control commands (start, stop, reference speed)
- These assemblies are linked using implicit messaging (real-time I/O data).

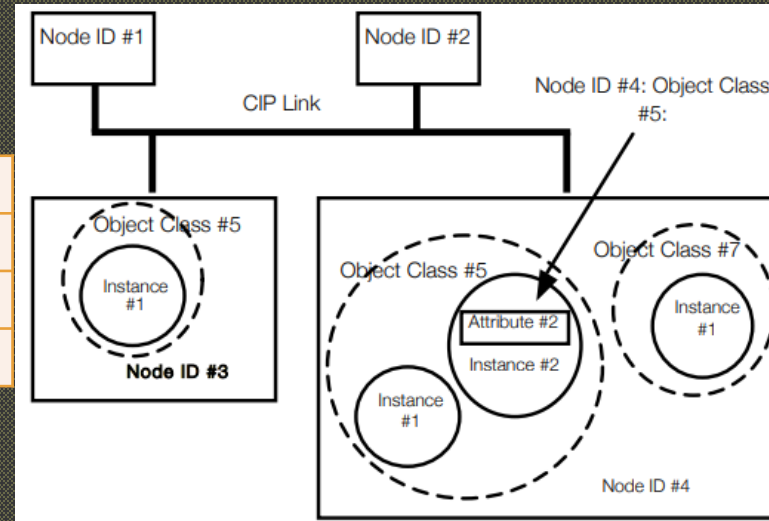
4. Connection Object

- The Connection Object (Class 0x05) manages the communication link.
 - Defines update rate (e.g., 10 ms)
 - Keeps the link alive (heartbeat)
 - Defines what happens if communication fails (safe state)

7B.4 CIP Object Hierarchy

Every EtherNet/IP device has:

Level	Meaning	Example
Class	Type of object	Identity Object
Instance	Specific copy of the object	Instance 1 = this device's identity
Attribute	Specific piece of data	Attribute 3 = Vendor ID



CIP Device (e.g., PowerFlex Drive)

- Class 0x01: Identity Object
 - Instance 1
 - Attributes: VendorID, SerialNumber, Revision
 - Services: Get_Attribute_Single
- Class 0x04: Assembly Object
 - Instance 100: Output Assembly (PLC → Drive)
 - Instance 101: Input Assembly (Drive → PLC)
- Class 0x0F: Parameter Object
 - Instance 1: MaxSpeed
 - Instance 2: AccelTime
 - Instance 3: ControlMode
- Class 0x05: Connection Object
 - Defines link between PLC and Drive
 - Uses implicit or explicit messaging

In CIP, every device is described using a set of objects. So, to read “Vendor ID”:

- Class = 1 (Identity Object)
- Instance = 1 (This device)
- Attribute = 1 (Vendor ID)

This is like saying: Folder = Identity Object; Then inside this Folder: File = Instance 1; Then in the file: Data = Vendor ID

- Each folder = Class (e.g., Identity Object, Assembly Object)
 - Each file inside that folder = Instance
 - Each file has attributes (data), services (commands you can send), and connections (communication links)

7B.4 CIP Object Hierarchy

CIP Object Hierarchy

The Common Industrial Protocol (CIP) organizes all data in a device into a structured, object-oriented model. This diagram shows the hierarchy from the device down to its individual data points (attributes).



Level	Meaning
Class	Type of object
Instance	Specific copy of the object
Attribute	Specific piece of data

7B.5 EtherNet/IP Communication Types

CIP supports two fundamental types of communication:

Type	Description	Transport Layer	Example
Explicit Messaging	Used for non-time-critical data exchange. Involves request/response transactions between devices.	TCP (Port 44818)	Reading a device parameter, changing configuration
Implicit Messaging	Used for real-time I/O data exchange. Data is sent cyclically at a specified update rate (RPI).	UDP (Port 2222)	Transmitting sensor or actuator data in control loops

Explicit Messaging

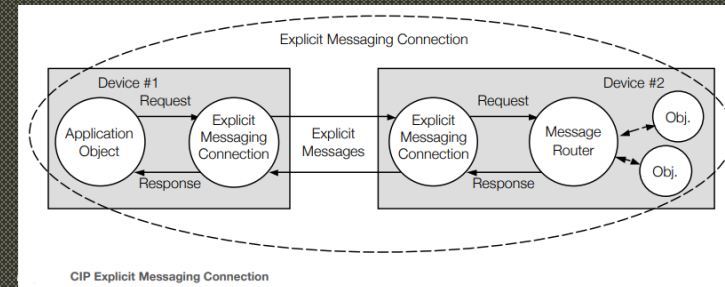
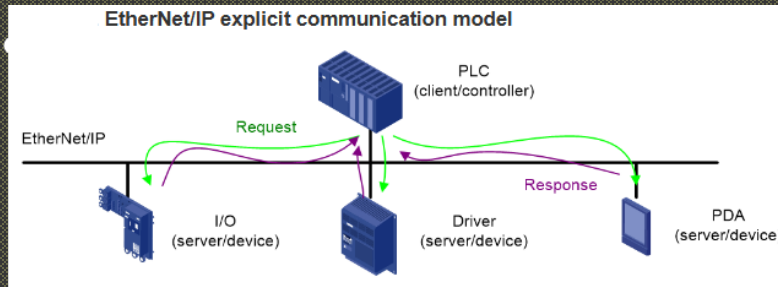
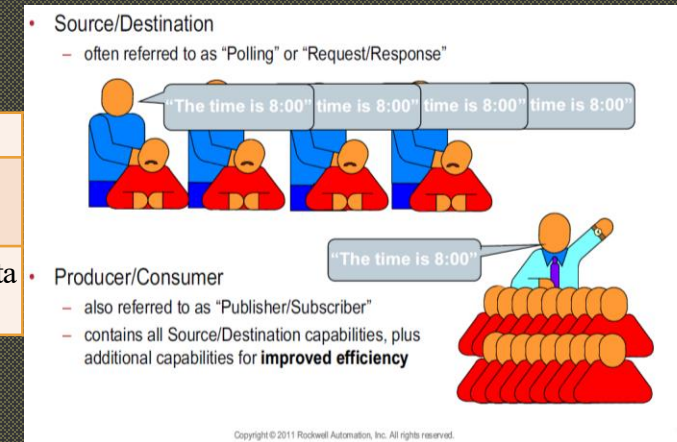
Explicit messaging works like a conversation between two devices:

- The client (e.g., PLC) sends a request message to the server (e.g., drive)
- The server processes the request and sends a response
- It includes addressing (where to go) and service information (what to do).
- The message tells the device exactly which object, which attribute, and which action (read, write, reset, etc.) to perform.
- This type of communication is typically used for configuration, diagnostics, or non-time-critical data, and it uses TCP on EtherNet/IP.

Example:

A PLC sends a message to a drive asking:

- “Read Attribute 3 (Firmware Revision) of the Identity Object.”



7B.5 EtherNet/IP Communication Types

Example: Reading Data via Explicit Messaging in Studio 5000

Step 1: Configure the Device

- In the I/O tree, add the EtherNet/IP device with its IP address.
- Ensure it supports the desired object class and attribute.

Step 2: Create the Message Instruction

- Insert a MSG instruction in your ladder logic.
- Set the Message Type to “CIP Generic.”
- Choose “Get Attribute Single.”
- Define:

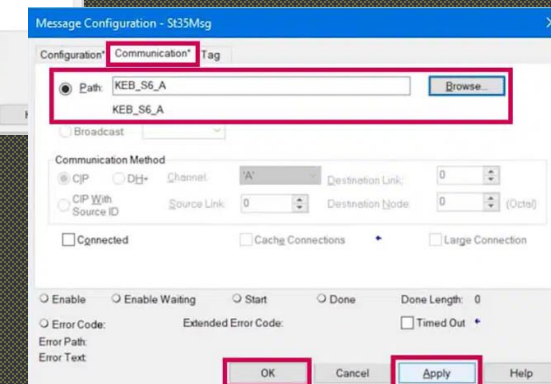
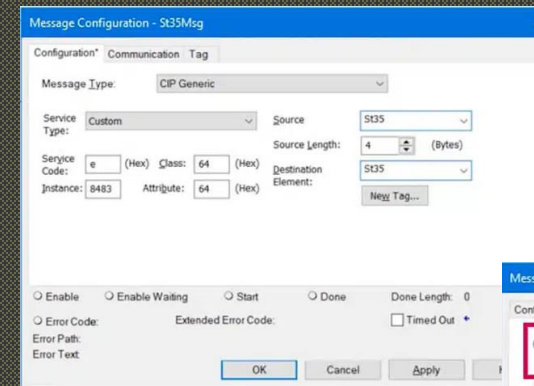
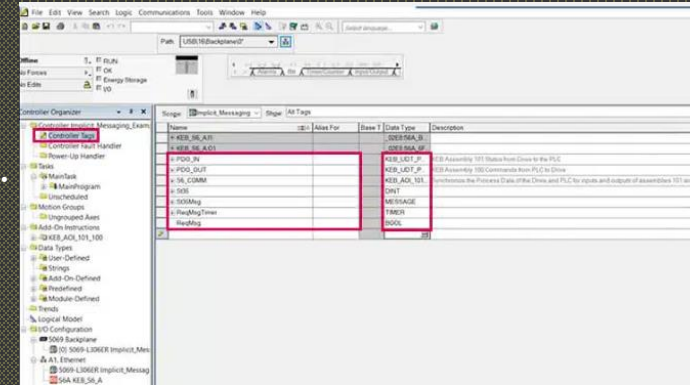
- Class:
- Instance:
- Attribute:

- Specify the destination tag to store the response.

Step 3: Trigger the Message

When the MSG instruction is executed, the PLC will:

- Open a TCP connection to port 44818 on the device.
- Send the CIP request.
- Wait for a response.
- Store the result in the destination tag.



7B.5 EtherNet/IP Communication Types

Implicit Messaging

Implicit messaging is used for continuous, time-critical I/O data and does not include specific address or service details. It operates on a producer/consumer model: one device produces data, and one or more devices consume it.

- Data is sent at a fixed interval called RPI (Requested Packet Interval).
- It uses UDP for faster, connectionless transmission.
- The message is identified only by a Connection ID, and the meaning of the data is implied by that ID, so no request/response is needed.

Example:

In a ControlLogix system:

- The PLC (scanner) produces output data and consumes input data.
- The EtherNet/IP I/O module (adapter) produces input data and consumes output data.

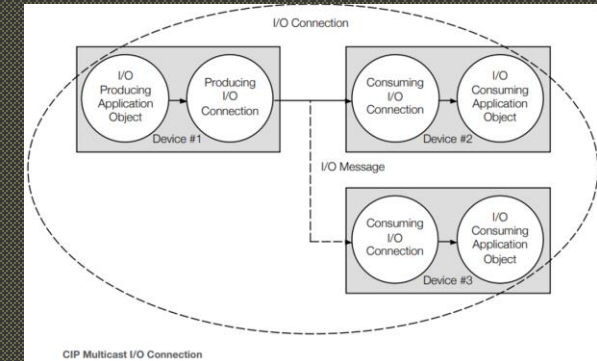
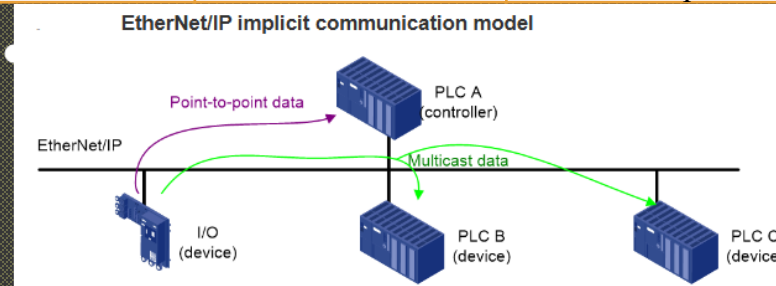
This exchange happens automatically once the connection is established.

A PLC and a sensor are already connected.

Every 10 ms, the sensor sends a data packet labeled with Connection ID “1234.”

The PLC knows that this packet always contains the current temperature value.

Type	Description	Transport Layer	Example
Explicit Messaging	Used for non-time-critical data exchange. Involves request/response transactions between devices.	TCP (Port 44818)	Reading a device parameter, changing configuration
Implicit Messaging	Used for real-time I/O data exchange. Data is sent cyclically at a specified update rate (RPI).	UDP (Port 2222)	Transmitting sensor or actuator data in control loops

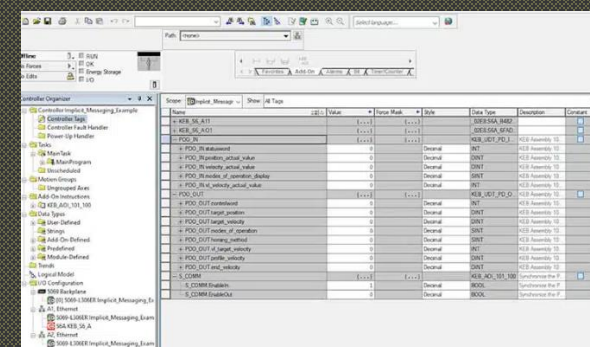
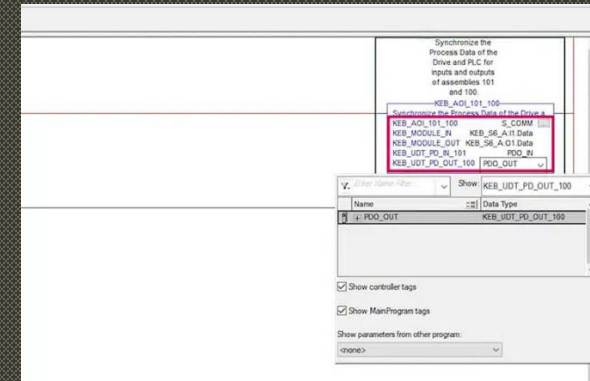
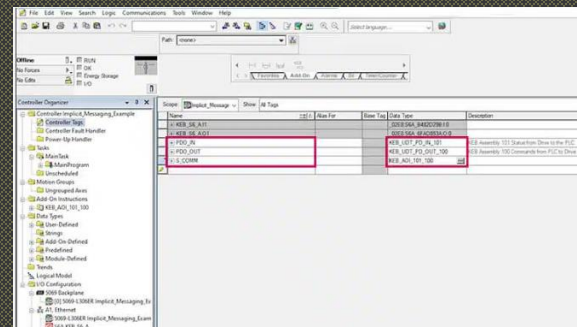
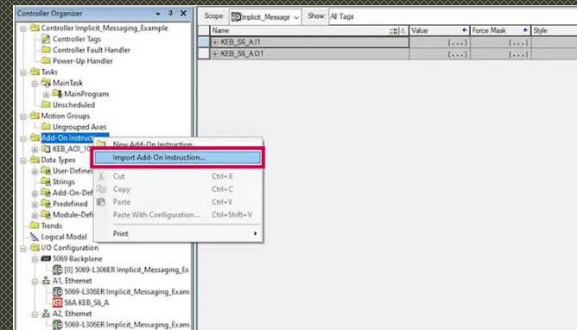
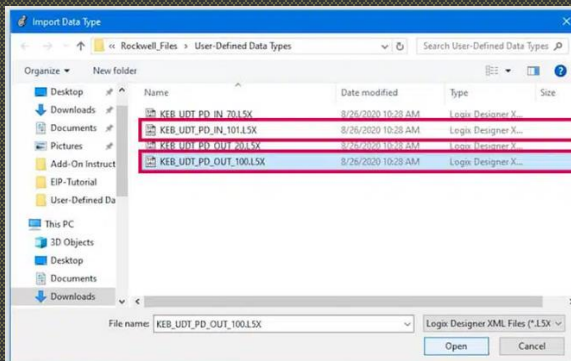
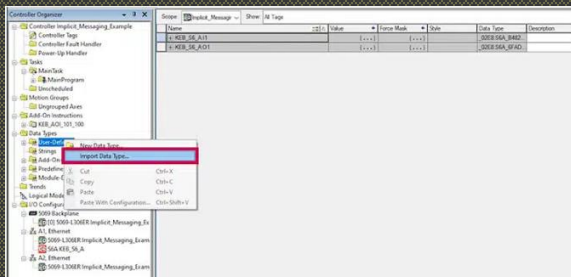


7B.5 EtherNet/IP Communication Types

When adding a remote EtherNet/IP adapter in Studio 5000:

- In the I/O Configuration, right-click and choose “New Module”.
- Select the correct device catalog (e.g., 1756-EN2T or 1734-AENT).
- Enter the IP address and RPI (e.g., 10 ms).
- Define the I/O data tags (e.g., Local:1:I.Data[0], Local:1:O.Data[0]).

Once downloaded, the PLC automatically exchanges input/output data cyclically via UDP, with no user code required.



7B.5 EtherNet/IP Communication Types

In CIP (Common Industrial Protocol), when a device (like a PLC) and another device (like a motor drive or sensor) set up an implicit connection, they agree on how to exchange data, and that connection is identified by a unique number called the Connection ID.

So, the Connection ID is just a special label (a unique number) that identifies one communication link between two devices.

Step 1 - Setting Up the Connection

- When configuring a drive in Studio 5000, the PLC and drive automatically exchange setup messages to create an I/O connection.

They agree on:

- What data will be sent (e.g., speed, status, commands)
- How often it will be sent (e.g., every 10 ms)
- The direction (PLC → Drive and Drive → PLC)
- And they assign Connection IDs

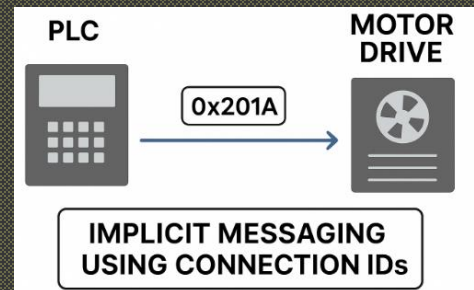
The receiving device sees the Connection ID and instantly knows:

- Who sent the data
- What kind of data it is (for example, “this is motor speed” or “this is drive status”)
- What to do with that data (for example, “update the output tag”)

Direction	Connection ID	Data Meaning
PLC → Drive	0x201A	Output data: Start command, Speed reference
Drive → PLC	0x202B	Input data: Actual speed, Status word

Step 2 - Data Exchange

- Now, every 10 ms:
- The PLC sends a UDP packet labeled 0x201A.
The drive sees “0x201A” and knows this data contains control commands.
- The drive sends a UDP packet labeled 0x202B.
The PLC sees “0x202B” and knows this data contains status and speed feedback.

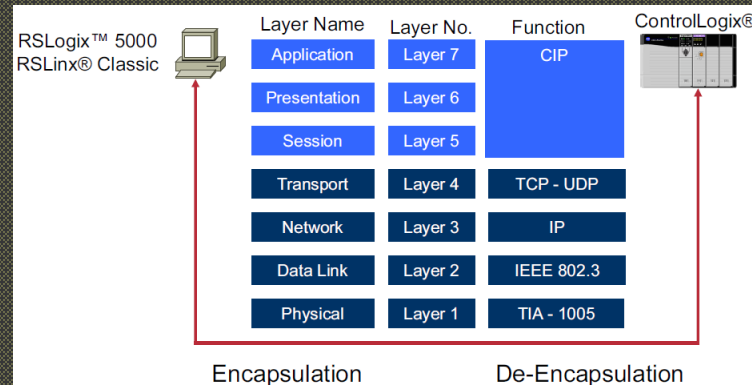
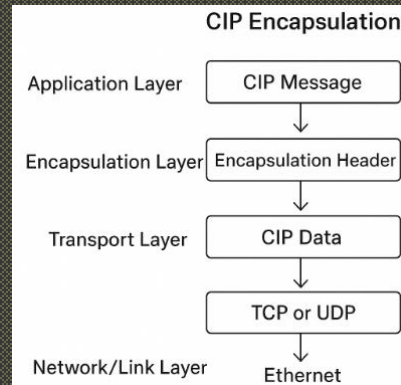


7B.6 CIP Encapsulation

CIP Encapsulation is the process of wrapping (or packaging) a Common Industrial Protocol (CIP) message inside a TCP or UDP packet so it can travel over a standard Ethernet network.

You can think of it like mailing a letter:

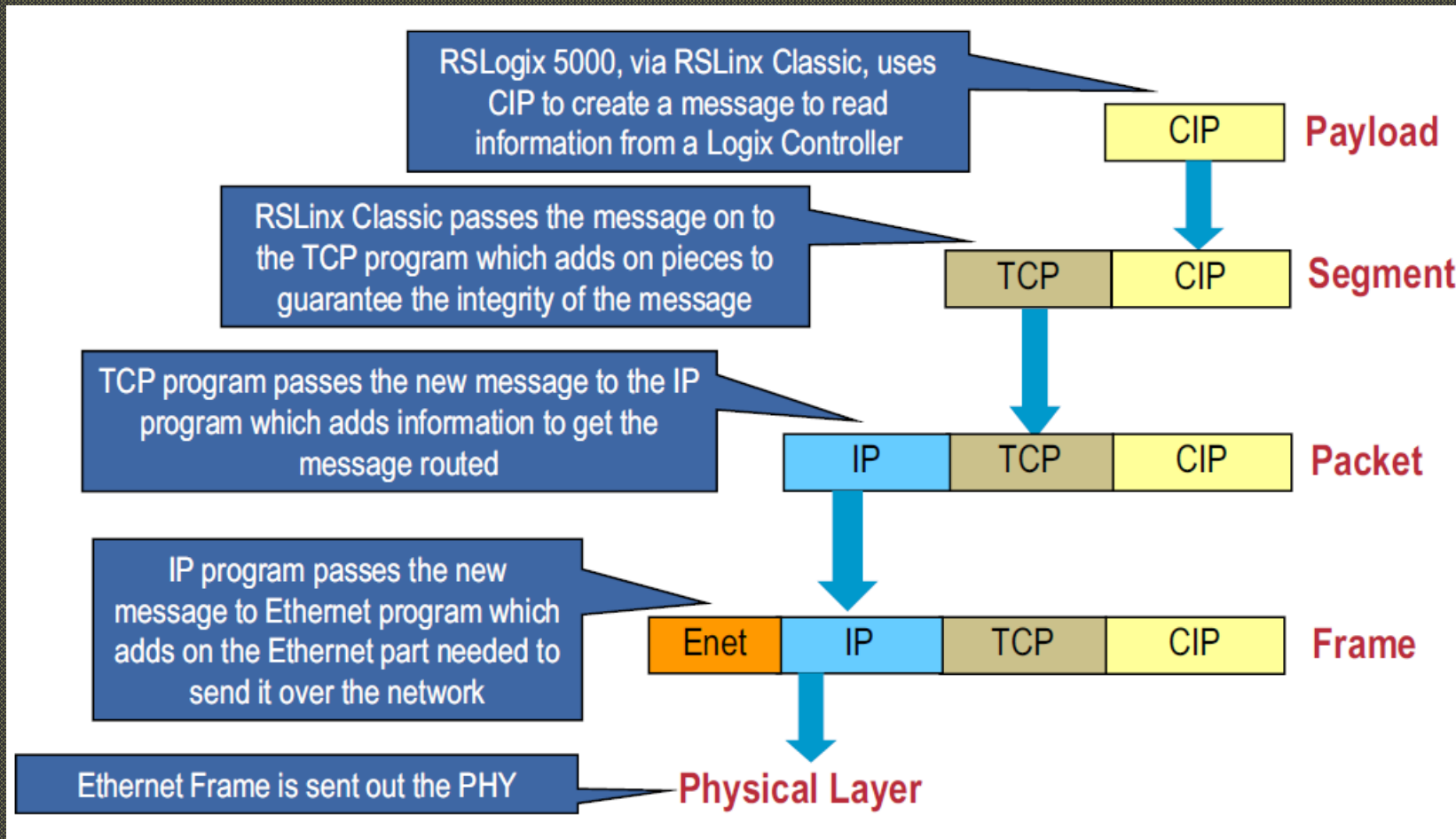
- The CIP message is the letter (the actual industrial data or command).
- The encapsulation header is the envelope, it provides addressing and control information so the receiver knows how to read and deliver the message.
- The TCP or UDP packet is the postal system, it carries the envelope (encapsulated message) from one device to another over Ethernet.



Fundamentals of EtherNet/IP Networking – Rockwell Automation 2011

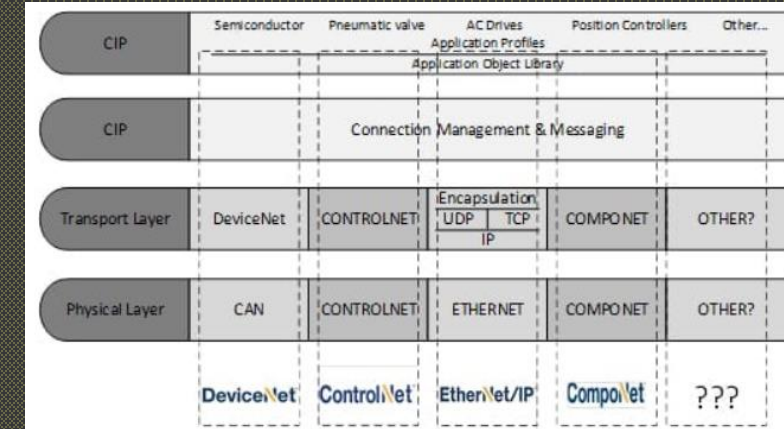
- CIP was designed to work across multiple network types (EtherNet/IP, DeviceNet, and ControlNet).
- Each physical network has its own way of sending data, so CIP uses encapsulation to adapt to the transport medium.
- In EtherNet/IP, messages must be sent over TCP/IP or UDP/IP, which are standard Internet protocols.

7B.6 CIP Encapsulation

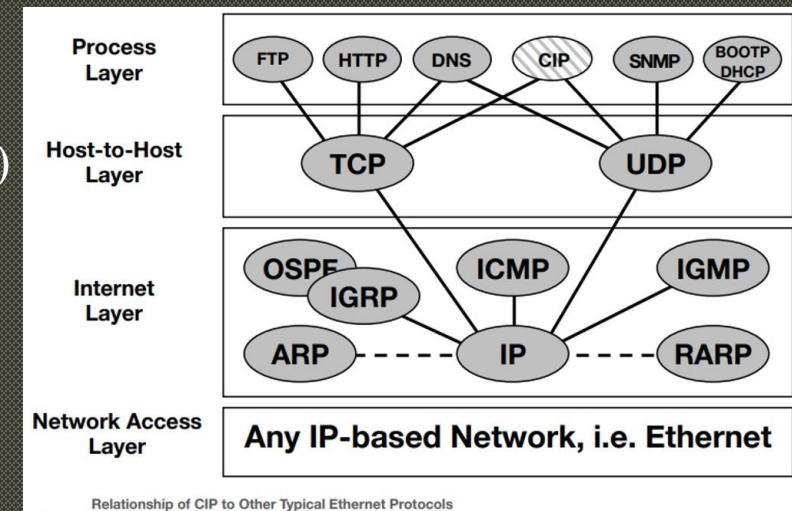


7B.6 CIP Encapsulation

- Configuration with EDS Files
 - Devices include a file called EDS (Electronic Data Sheet).
 - It tells configuration software how to set up the device, including:
 - What connections are possible
 - Parameters and settings to be used
- Conformance Testing
 - Devices are tested to make sure they work correctly with others on an EtherNet/IP network.
- TCP/IP Requirements
 - EtherNet/IP runs on standard Internet protocols:
 - IP, TCP, UDP, ARP, ICMP, IGMP, Ethernet (IEEE 802.3)
 - These protocols let devices send and receive messages properly.
- Working with Other Protocols
 - EtherNet/IP can work alongside other Ethernet protocols like:
 - HTTP (web access), FTP (file transfer), Telnet (remote login)
 - It fits well into networks that already use these services.



<https://www.rtautomation.com/rtas-blog/what-is-cip/>



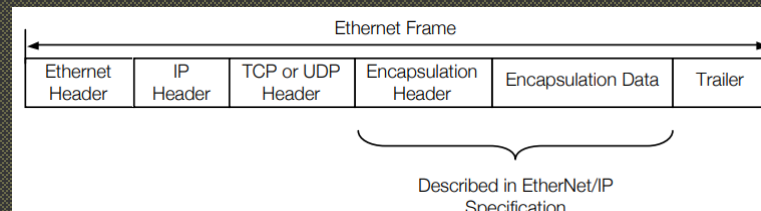
7B.6 CIP Encapsulation

EtherNet/IP uses standard TCP/IP and UDP/IP protocols without modification:

- TCP/IP (port 0xAF12) is used for explicit messaging and for encapsulation protocol commands during communication setup.
- UDP/IP (port 0x08AE) is used for I/O messaging, which supports unicast and multicast.
- The encapsulation header includes a command that defines the meaning of the encapsulated data.
- Many commands use the Common Packet Format (CPF: Header - Payload - Trailer) to structure the data.
- While UDP-based I/O messages do not include an encapsulation header, they still conform to the CPF structure.

Each encapsulated message includes two main parts:

- Encapsulation Header (24 bytes)
 - Tells the receiving device what kind of command it is (e.g., “SendRRData” or “RegisterSession”)
 - Contains session information and message length.
- CIP Data (variable length)
 - The actual CIP message: could be a request, a response, or an I/O data exchange.



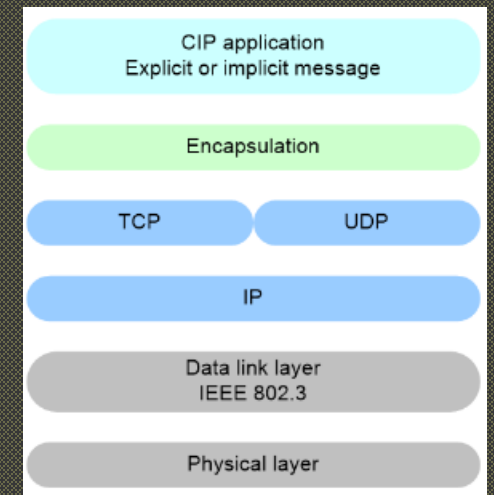
7B.6 CIP Encapsulation

Step 1: Session Registration

- Before anything happens, both devices must establish a session.
- The PLC sends a RegisterSession command.
- The adapter (I/O device) replies with a session handle (a unique number for this communication).
- Think of this as both devices agreeing to “start a conversation.”

Step 2: Send a CIP Message

- Once the session exists, the PLC can send any CIP message, for example:
- “Read Attribute” from the Identity Object (0x01).
- “Set Attribute” to change a parameter in the Assembly Object (0x04).
- This is wrapped in an encapsulation command called:
- SendRRData (for Explicit Messages), it uses TCP.
- SendUnitData (for Implicit I/O Messages), it uses UDP.

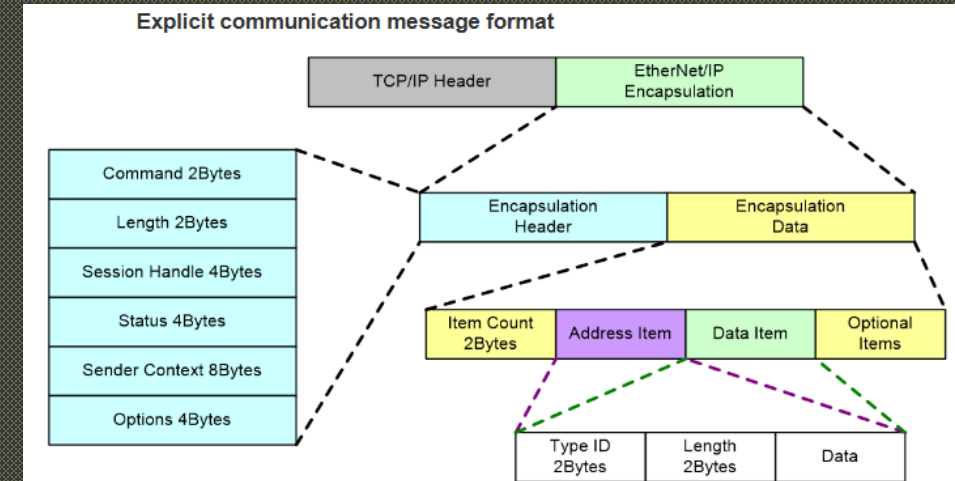
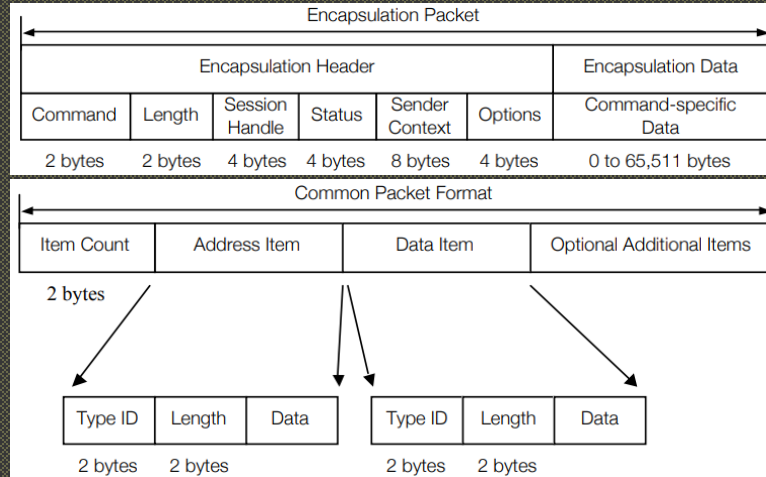


https://www.h3c.com/en/d_202408/2239284_294551_0.htm

Step 3: Receive and Process

- The receiving device removes (decapsulates) the Ethernet and TCP/UDP headers, then the encapsulation header, and finally reads the actual CIP message.
- It processes the request (e.g., read an attribute) and sends back an encapsulated response.

7B.6 CIP Encapsulation



https://www.h3c.com/en/d_202408/2239284_294551_0.htm

Enc. Header	Size (bytes)	Description
Command	2	Specifies the EtherNet/IP command to be executed (e.g., 0x006F = SendRRData, 0x006E = RegisterSession).
Length	2	Number of bytes of data that follow this header (the length of the encapsulated command-specific data).
Session Handle	4	Identifies an active communication session between two devices (established using the RegisterSession command).
Status	4	Indicates whether the previous command was executed successfully (0x0000 = success).
Sender Context	8	Optional user-defined field; often used by the sender to match requests and responses.
Options	4	Reserved for future use; typically set to 0x00000000.

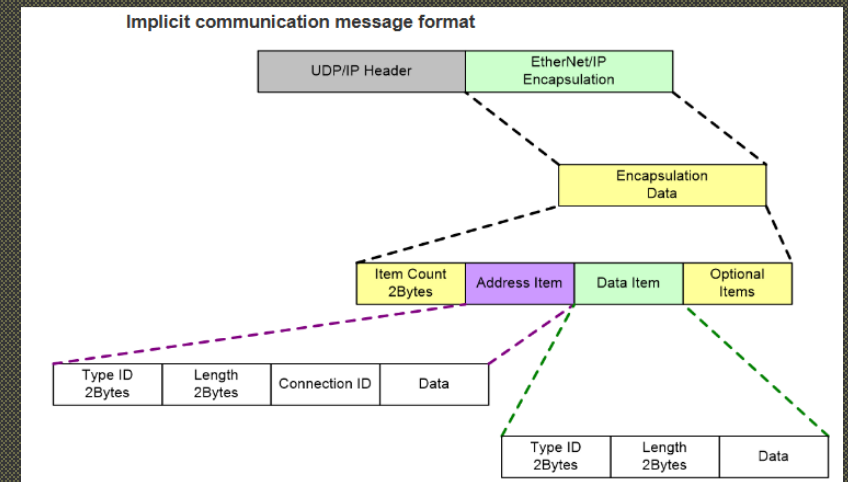
Enc. Data	Description
Item Count	Indicates how many “items” (pieces of data) are included in this packet. For most messages, this value is 2 (address item + data item).
Address Item	Contains information about the destination. Usually set to indicate a NULL address item (0x0000) for explicit messages (because TCP already manages addressing).
Data Item	Contains the actual CIP message data, which includes:
Data Type	Identifies the type of item (e.g., “connected” or “unconnected” message).
Data Length	Length of the CIP data in bytes.
CIP Message	The actual CIP command and path (e.g., “Get Attribute Single” request to Class 0x01, Instance 1, Attribute 1).

7B.6 CIP Encapsulation

The EtherNet/IP implicit encapsulation packet (used for real-time I/O messaging) does not include the encapsulation header like in explicit messaging.

Instead, it includes only the encapsulation data, which carries the information needed for fast cyclic data exchange. This is transmitted using the SendUnitData command over UDP.

Field	Description
Item Count	Indicates how many items are included in this data packet. For implicit messaging, the value is typically 2 — one address item and one data item.
Address Item	Contains the CIP Connection ID, which identifies the specific I/O connection. In implicit messaging, the Connection ID determines the destination and meaning of the data. The address item also includes the address type and length fields.
Data Item	Contains the actual I/O data being transferred. This includes the data type, data length, and the payload (the input or output data for the connection).



https://www.h3c.com/en/d_202408/2239284_294551_0.htm

Feature	Explicit Messaging	Implicit Messaging
Command	SendRRData (TCP)	SendUnitData (UDP)
Includes Header?	Yes	No
Addressing	Object, Class, Instance, Attribute	Connection ID
Data Meaning	Defined by message content	Implied by connection setup
Usage	Configuration, diagnostics	Real-time I/O data
Speed	Slower (TCP)	Fast (UDP, cyclic)

7B.7 EtherNet/IP – Topologies

Common Topologies

- Primarily uses star topology with Layer 2/3 switches.
- Can also use linear, ring, or branched topologies.

Star Topology:

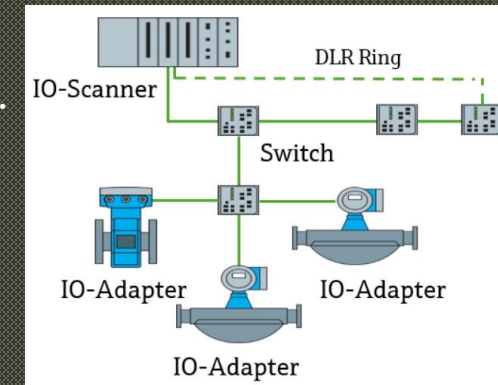
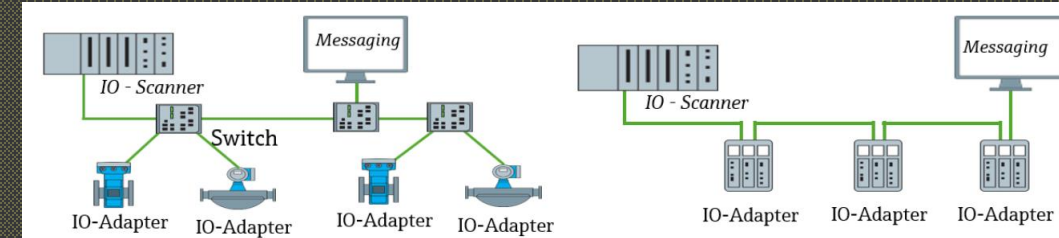
- All devices (sensors, actuators, controllers) connect to a central switch or interconnected switches.
- Each device uses a dedicated point-to-point link to the switch.

Key Advantages:

- Simplified Troubleshooting: Failures are isolated and don't affect the whole network.
- High Data Rates: Supports the bandwidth needed for real-time control.
- Scalability: Easy to add devices - just plug into the switch.
- Resilient: More robust than bus or ring topologies, though the switch remains a single point of failure.

Linear Topology

- In a linear topology, devices are connected one after another in a straight line.
- Each device has two Ethernet ports, acting like a mini switch.
- If one device or cable fails, communication is lost beyond that point.
- This setup is simple but not fault-tolerant.



<https://netilion.endress.com/blog/ethernet-ip/>



<https://www.rtautomation.com/rtaas-blog/are-you-dlr-ready/>

7B.7 EtherNet/IP – Topologies

Ring Topology with Device Level Ring (DLR)

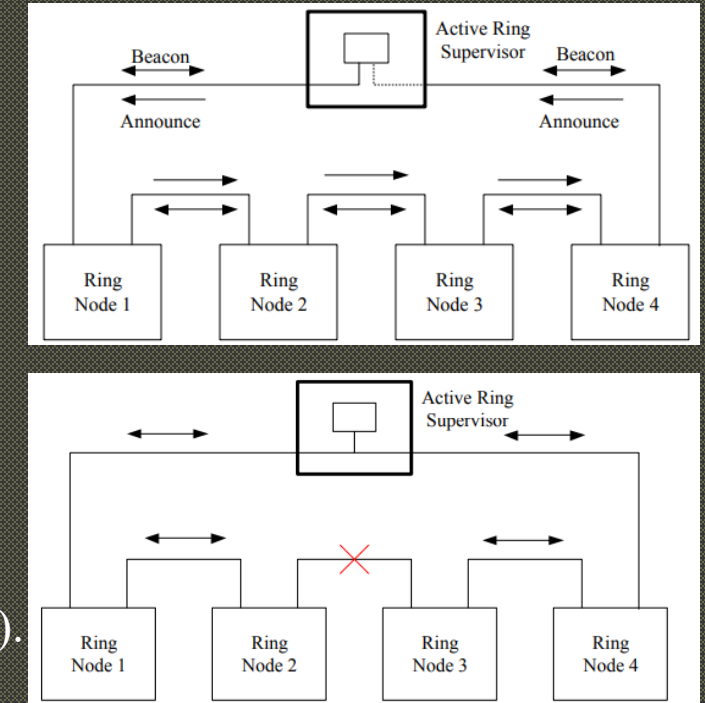
- Device Level Ring (DLR) is an Ethernet-based redundancy protocol used in industrial automation networks to maintain continuous communication even if a cable or device fails.

Key Features

- Creates a ring topology at the Ethernet (Layer 2) level.
- Provides fast fault recovery, typically under 3 ms for up to 50 devices.
- Operates below TCP/IP and CIP, completely transparent to higher protocols.
- Automatically detects and corrects faults without manual intervention.

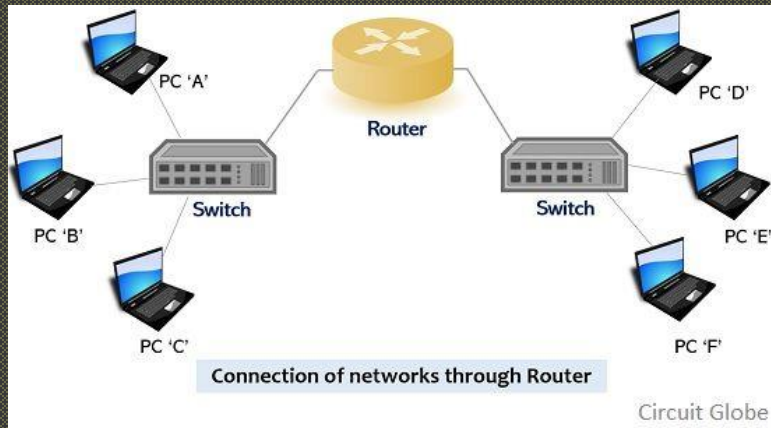
How It Works

- The network forms a closed ring between devices (each device has two Ethernet ports).
- One device acts as the Ring Supervisor, all others are Ring Nodes.
- The supervisor sends Beacon frames (every 400 μ s) to monitor ring health.
- If a break is detected (missing Beacons), traffic is rerouted instantly using the alternate path.
- When the fault is fixed, the ring automatically restores normal operation.

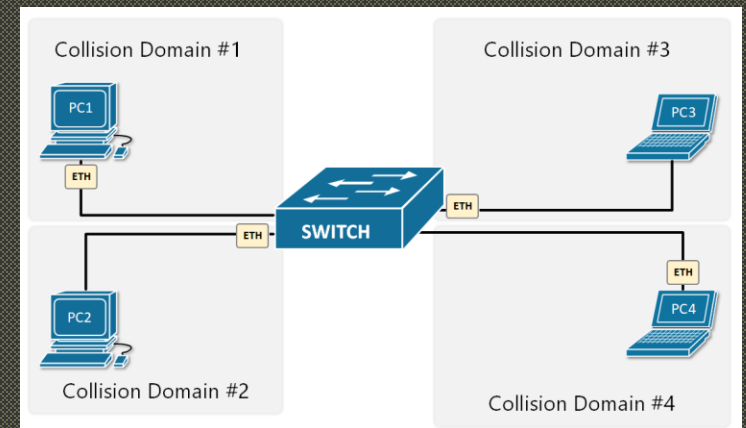
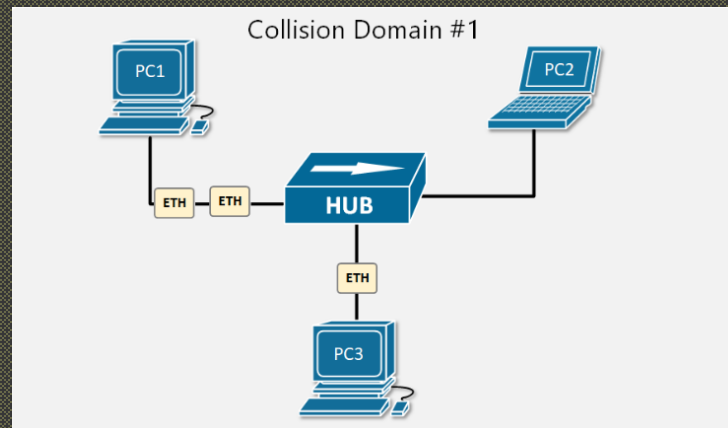


7B.7 EtherNet/IP – Topologies

- Ethernet Network Setup
 - For reliable and real-time communication, avoid using old hubs.
 - Use network switches to avoid message collisions.
 - Use routers to separate industrial devices from office computers - but still allow monitoring and setup from PCs.
- Devices with Multiple Ethernet Ports
 - Some devices have more than one Ethernet port.
 - The CIP specification explains how to set up these devices correctly, for different topologies (ring, line, etc.).



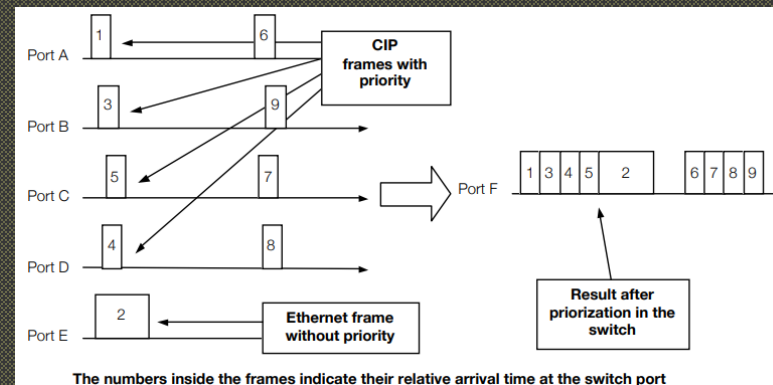
https://circuitglobe.com/difference-between-router-and-switch.html#google_vignette



<https://www.networkacademy.io/ccna/ethernet/collision-domains>

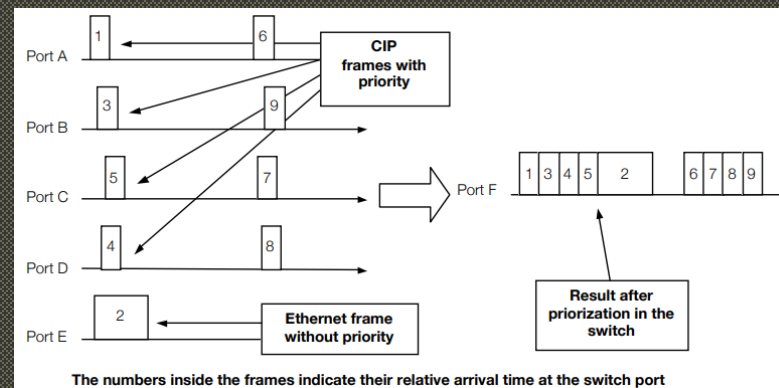
7B.8 Message Prioritization

- Think of an Industrial Network as a Highway
- Fast Cars & Motorcycles = Critical Data: Sensor readings and control signals (like "STOP the machine!") that need to get through immediately.
- Big Trucks = Normal Data: Large file transfers, web traffic, or emails. Important, but not time-sensitive.
- The Problem: Traffic Jams
- If all vehicles share the same lane, a slow-moving truck can block fast cars and motorcycles. In an industrial network, this means critical messages can be delayed, potentially causing machines to behave incorrectly or dangerously.
- The Solution: QoS (Quality of Service) - The Express Lane
- QoS is like creating priority lanes on the highway. It lets the network decide:
 - High-priority data (critical messages) go straight through.
 - Low-priority data (big files) wait their turn.



7B.8 Message Prioritization

- How the Network Knows Who Goes First
 - DSCP Tag = Shipping Label:
Each data packet gets a “priority label” (like “URGENT” or “STANDARD”) in its header. Routers read this label to decide which packets to send first.
 - 802.1Q Tag = VIP Pass:
A tag added directly to the Ethernet frame. Switches see it and let high-priority data jump ahead. It also tells the switch which “private highway” (VLAN) the data belongs to.
- Special Case: Precision Time Protocol (PTP)
 - PTP is like an atomic clock for the network.
 - It synchronizes all devices with sub-microsecond accuracy.
 - PTP messages get the highest priority to ensure precise timing.



7B.9 Physical Layer

- Industrial Suitability and Physical Layer Requirements:
- EtherNet/IP is adapted for factory environments with two performance levels:
 - COTS (Commercial Off-The-Shelf): Basic Ethernet functionality using standard RJ-45 connectors, suited for IP20-rated applications, limited to 100 meters using Cat 5e/6 cables.
 - Industrial Level: Enhanced durability with sealed RJ-45 or M12 connectors, compliant with IP67 standards, and designed for harsh environments.
- Cabling and Media:
 - Cat 5e/6 cables (shielded/unshielded) are standard.
 - Shielded cables are recommended near metal enclosures.
 - Fiber optic is preferred for long distances, high EMI environments, or high voltage isolation.
- Frame and Protocol:
 - Uses standard Ethernet frames with TCP/IP and UDP/IP encapsulation.
 - No special frame structure needed; fragmentation handled by IP layers.
 - Fully supports CIP features without scaling requirements.
- Indicators and Configuration:
 - Devices must include Module Status and Network Status indicators.
 - Configuration may be manual or automatic, depending on design.

"IP": stands for Ingress Protection.

"2": in IP20 indicates protection against solid objects larger than 12mm.

"0": in IP20 indicates no protection against liquids.

"6": in IP67 indicates highest level of protection against solid objects.

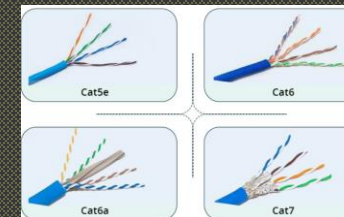
"7": in IP67 indicates level of protection against liquids - temporary submersion in water up to 1 meter deep for 30 minutes .

Cat6:

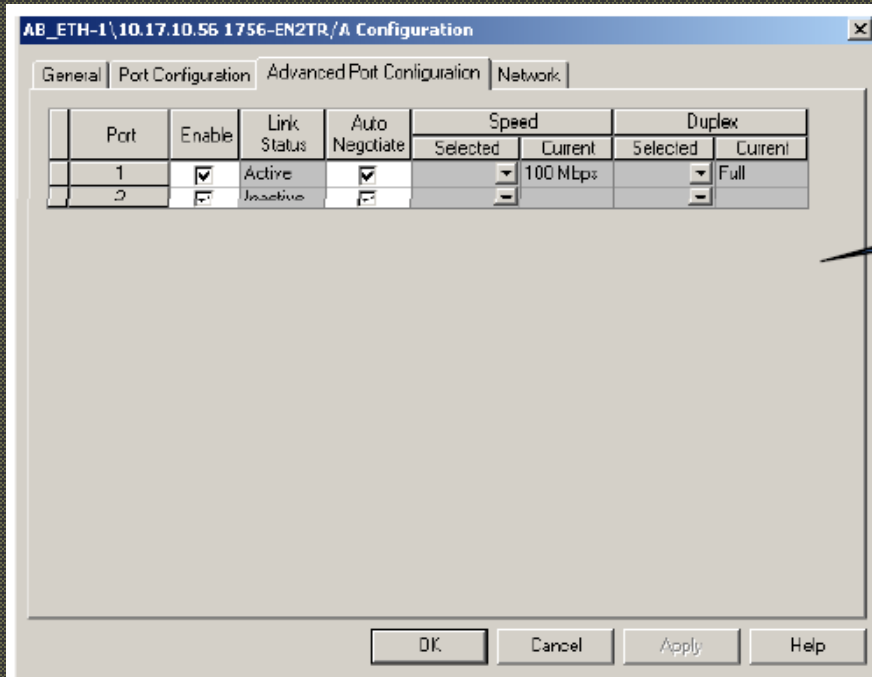
- Supports data transfer speeds up to 10 Gbps.
- Typically 250 MHz bandwidth.
- Better suited for demanding applications like large file transfers.
- Better resistance to EMI than Cat5e.

Cat5e:

- Supports data transfer speeds up to 1 Gbps.
- Typically 100 MHz bandwidth.
- Good for basic home and office networks, standard data transfer.



7B.9 Physical Layer

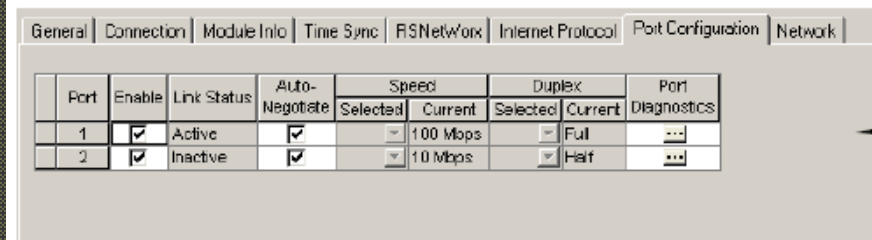


RSLogix 5000
Module Configuration

EN2TR Webpage
Network Settings

Ethernet Port 1

Interface State	Enabled
Link Status	Active
Speed	100 Mbps
Duplex	Full Duplex
Autonegotiate Status	Autonegotiate Speed and Duplex



RSLogix 5000
EN2TR Properties

7B.9 Physical Layer

MICE is a classification system used to describe the installation environment of industrial components in terms of four factors:

- M – Mechanical stress
- I – Ingress (dust, moisture)
- C – Climatic conditions (temperature, humidity)
- E – Electromagnetic interference (EMI)

Classification Levels - Each factor is rated from 1 (low) to 3 (high):

- M1I1C1E1 = Typical office environment
- M3I3C3E3 = Typical industrial environment

Not all environments fit neatly into one category. For example: An automation island may have: Strong mechanical shock (M3), Light dust (I2), High temperature (~65°C, C2), High EMI from welding robots (E3) → Combined classification: M3I2C2E3

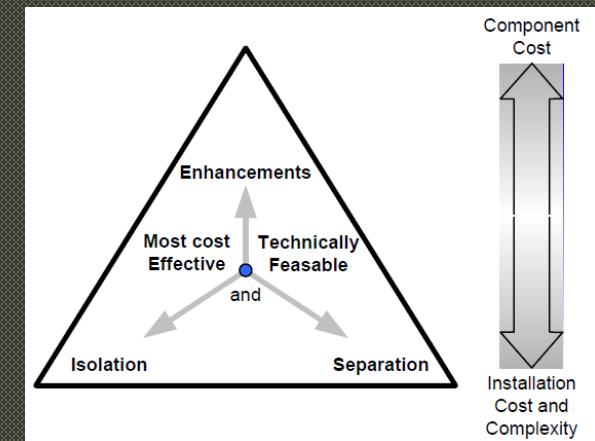
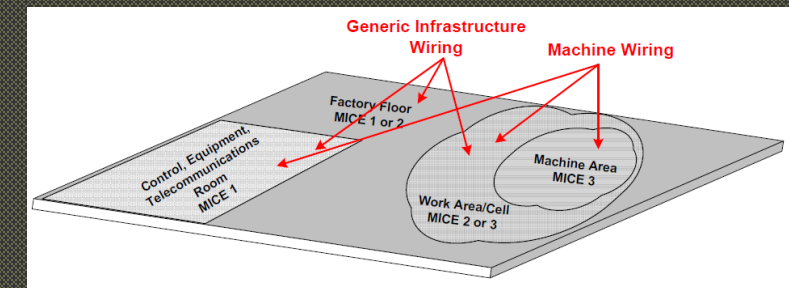
PurposeClassifying the environment using MICE helps:

- Select appropriate components
- Design robust cabling systems
- Decide if EMI, dust, or temperature mitigation is needed

This ensures better reliability and performance in harsh industrial settings.

Office	Lt. Industrial	Industrial
M1 I1 C1 E1	M2 I2 C2 E2	M3 I3 C3 E3

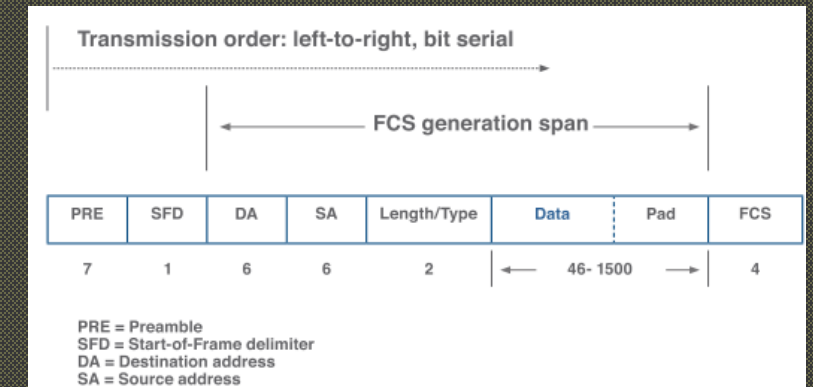
MICE classifications.



ODVA Media Planning and Installation Manual

7B.10 Data Link Layer

- Standard & Mechanism:
 - Uses IEEE 802.3 standard for data transmission.
 - Employs CSMA/CD (Carrier Sense Multiple Access/Collision Detection) for managing data access and collision response.
- Duplex Mode:
 - Originally half-duplex (send *or* receive), which caused delays.
 - Now operates in full-duplex, allowing simultaneous send and receive, eliminating traffic jams - ideal for real-time control.
- MAC Protocol:
 - Every device has a unique MAC address (6 bytes) used to identify the sender and recipient.
 - Multicast capability is supported via address formatting.
- Ethernet Frame:
 - Can carry up to 1,500 bytes of data.
 - Enables high-throughput communication for increasingly intelligent, compact, and cost-effective devices.
- Benefits for Industrial Use:
 - Combines standard Ethernet, TCP/IP, and UDP.
 - Supports easy configuration, reliable communication, and integration between the factory floor and enterprise systems.

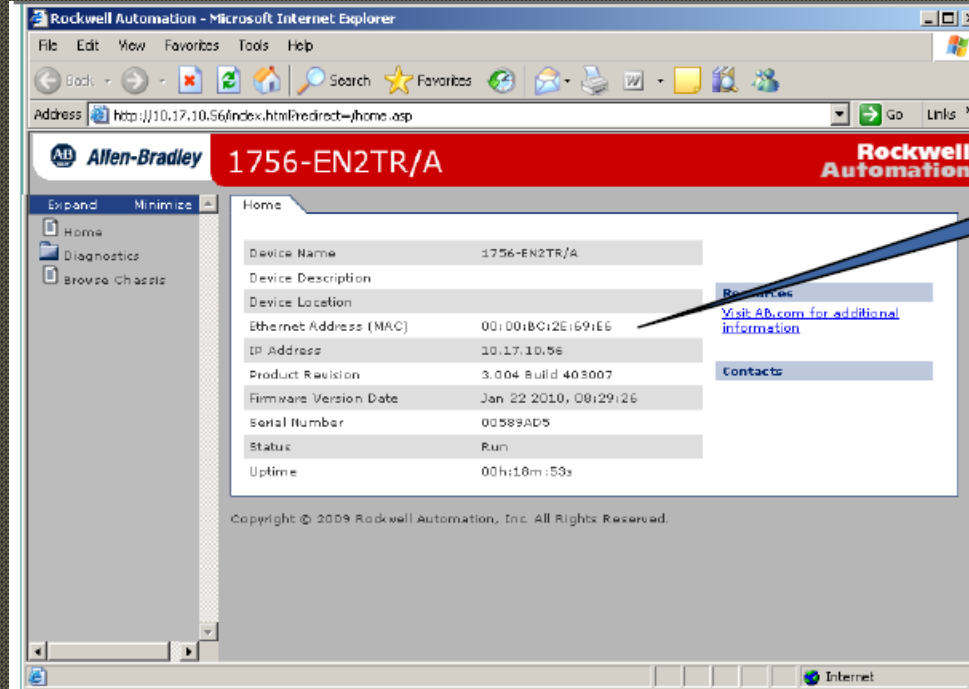


ODVA PUB00138R8

7B.10 Data Link Layer

Responsible for:

- Framing packets for transmission
- MAC addressing (Media Access Control) between devices
- Error detection using CRC (Cyclic Redundancy Check)



EN2TR Webpage
MAC Address

Media Counters Port 1

Alignment Errors	0
FCS Errors	0
Single Collisions	0
Multiple Collisions	0
SQE Test Errors	0
Deferred Transmissions	0
Late Collisions	0
Excessive Collisions	0
MAC Transmit Errors	0
Carrier Sense Errors	0
Frame Too Long	0
MAC Receive Errors	0

Media Counters:

Current Port: 1

Counter Name	Value	Counter Name	Value
Alignment Errors	0	Late Collisions	0
FCS Errors	0	Excessive Collisions	0
Single Collisions	0	MAC Transmit Errors	0
Multiple Collisions	0	Carrier Sense Errors	0
SQE Test Errors	0	Frame Too Long	0
Deferred Transmissions	0	MAC Receive Errors	0

RSLink Classic
EN2TR Diagnostics
Ethernet Statistics

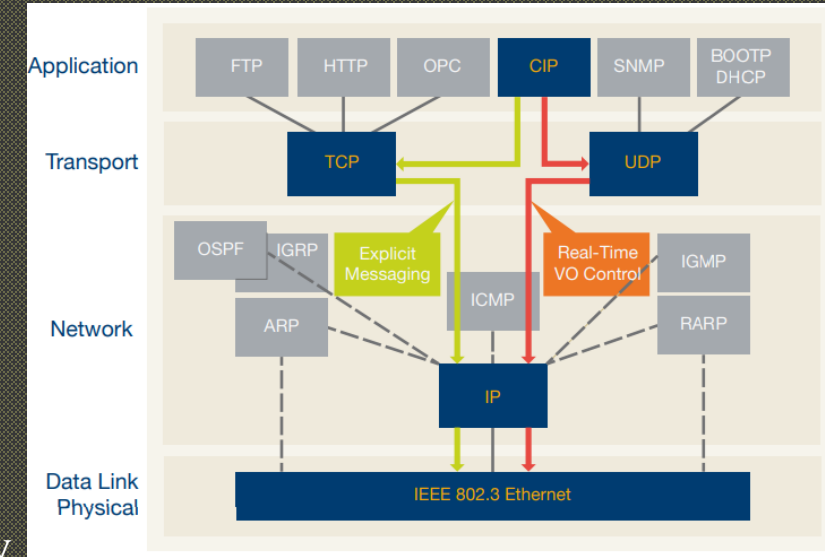
EN2TR Webpage
Ethernet Statistics

7B.11 Network and Transport Layers

EtherNet/IP uses standard TCP/IP and UDP/IP protocols at the Network and Transport Layers to ensure reliable communication across industrial networks.

Core Protocol Functions

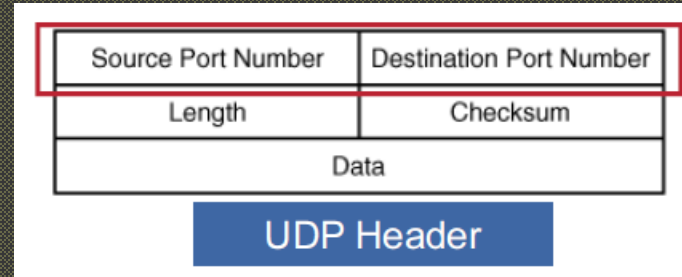
- TCP/IP (Transmission Control Protocol / Internet Protocol):
 - Ensures reliable, connection-oriented communication for configuration, diagnostics, and explicit messaging.
 - Provides data flow control, fragmentation reassembly, and message acknowledgments.
 - Ideal for client-server (explicit) communications (e.g., configuration or diagnostics).
 - Enables routing across multiple paths, offering robustness and interoperability.
- UDP/IP (User Datagram Protocol / Internet Protocol):
 - Used for real-time I/O (implicit) data via multicast, enabling producer-consumer communications.
 - Low overhead, connectionless, ideal for fast, time-critical control data.
 - Supported by timeout and watchdog mechanisms for reliability.
- CIP Encapsulation:
 - CIP (Common Industrial Protocol) messages are encapsulated in TCP or UDP packets, allowing seamless communication over standard Ethernet networks.



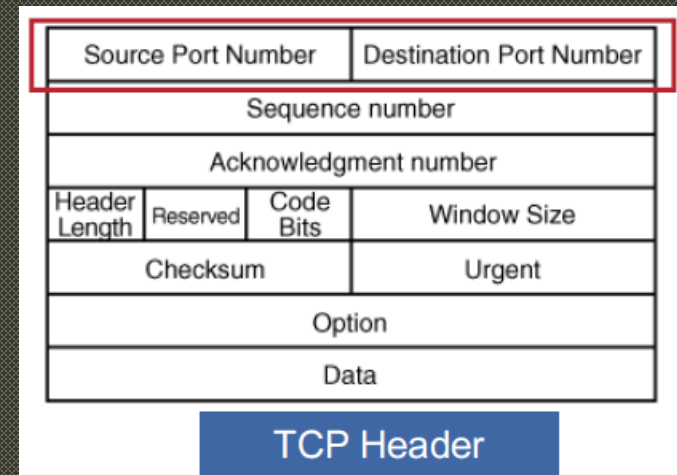
ODVA PUB00138R8

7B.11 Network and Transport Layers

- User Datagram Protocol - UDP
 - Connectionless/best effort
 - Does not use acknowledgements
 - IP - Unicast and Multicast
 - CIP - used for (Implicit) I/O - port 2222



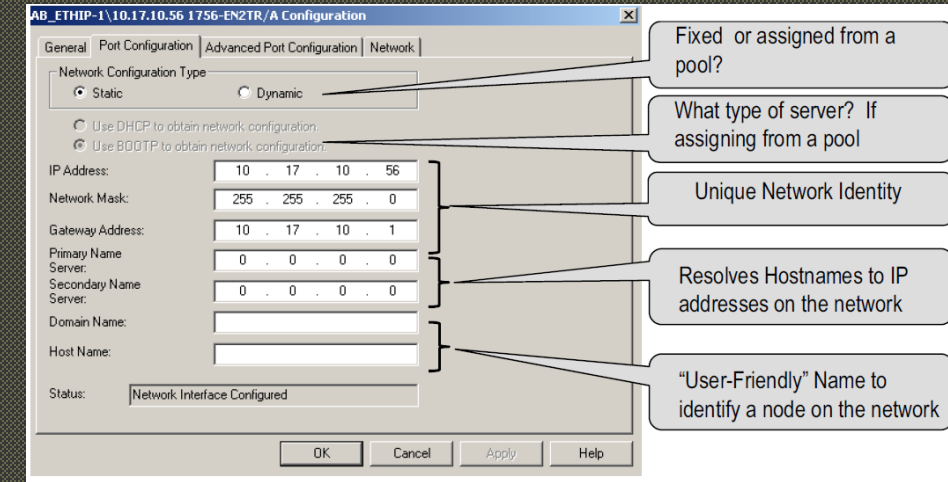
- Transmission Control Protocol - TCP
- Connection-oriented, end-to-end reliable transmission
- Utilizes acknowledgements (ACK) to ensure reliable delivery
- IP - Unicast
- CIP – used for (Explicit) messaging such as Operator Interface - port 44818



7B.11 Network and Transport Layers

Messaging Types in EtherNet/IP

- **Unconnected Messaging:**
 - Low-priority or one-time messages, managed by the Unconnected Message Manager (UCMM).
 - Uses TCP/IP.
- **Connected Messaging:**
 - Reserved communication channels for frequent or time-sensitive interactions.
 - Initiated by a Connection Originator, received by a Connection Target.
 - Two types:
 - **Explicit Messaging:** Point-to-point, client-server style. Uses TCP/IP.
 - **Implicit Messaging:** Real-time I/O data, often multicast. Uses UDP/IP.



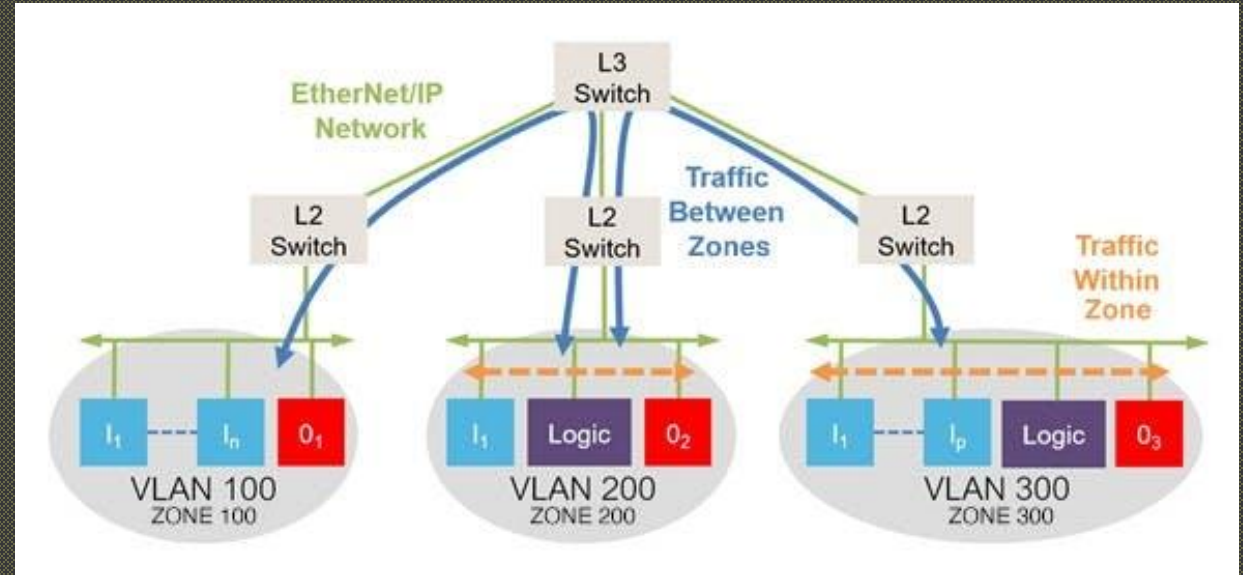
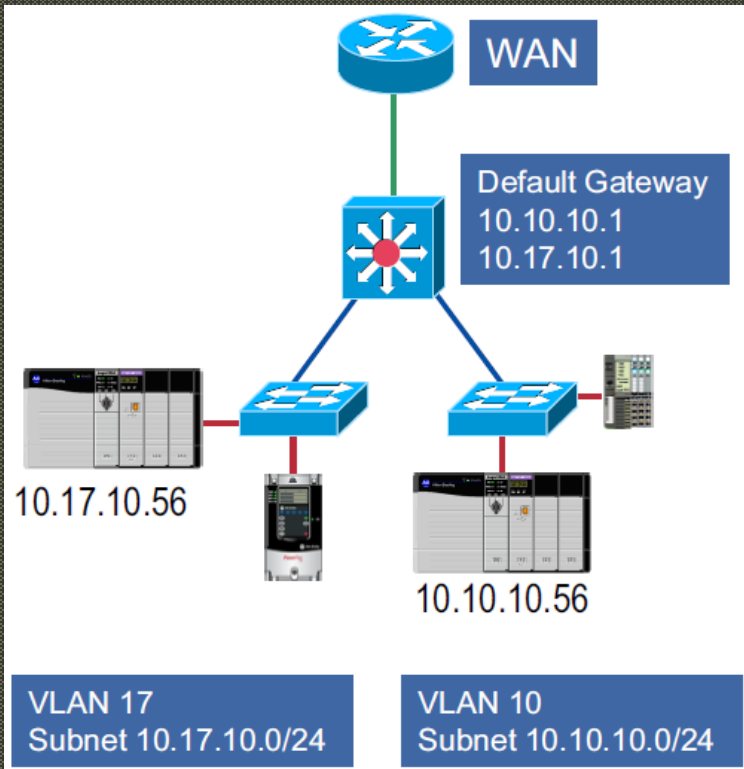
Fundamentals of EtherNet/IP Networking – Rockwell Automation 2011

Network Layer

- **Unconnected** = Occasional, slower, reliable.
- **Connected** = Continuous, faster, for control.
 - **Explicit** = Talk and wait for reply.
 - **Implicit** = Stream data quickly and constantly.

Type	Use Case	Protocol	Example	Analogy
Unconnected	Occasional, setup, or low-priority	TCP/IP	Read device info	Email
Explicit (Connected)	Configuration, commands	TCP/IP	Change motor speed	Phone call
Implicit (Connected)	Real-time control/data	UDP/IP	Continuous sensor updates	Walkie-talkie

7B.11 Network and Transport Layers



<https://www.automationworld.com/home/article/33001839/ethernet-ip-a-cip-based-protocol-for-discrete-and-process-applications>

Fundamentals of EtherNet/IP Networking – Rockwell Automation 2011

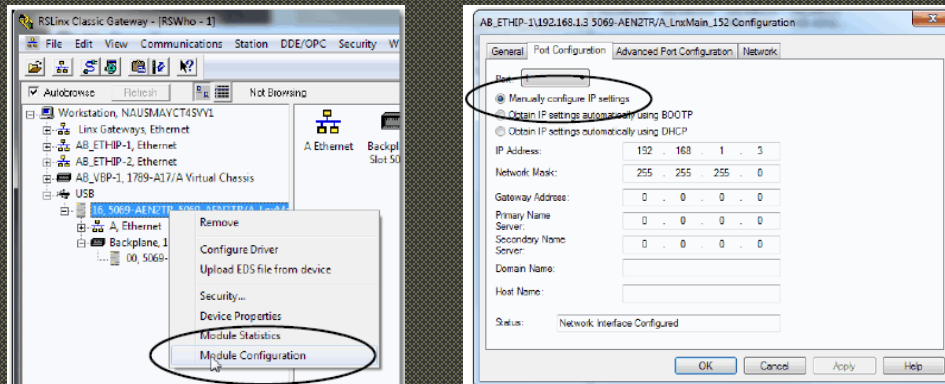
Routing Table	
Network	Port
10.17.10.0	1
10.10.10.0	2

7B.12 Rockwell Software

In industrial environments, unlike office settings, IP address consistency is critical. Devices like PLCs must retain the same IP address after each reboot to ensure reliable communication, unlike PCs that can use DHCP to receive dynamic addresses. Because Ethernet is the only universal interface on EtherNet/IP devices, IP configuration must be possible via Ethernet. The TCP/IP Interface Object in the EtherNet/IP specification supports several methods for IP assignment:

- BOOTP
- DHCP
- Explicit messaging (using Set_Attribute_Single or Set_Attributes_All services)

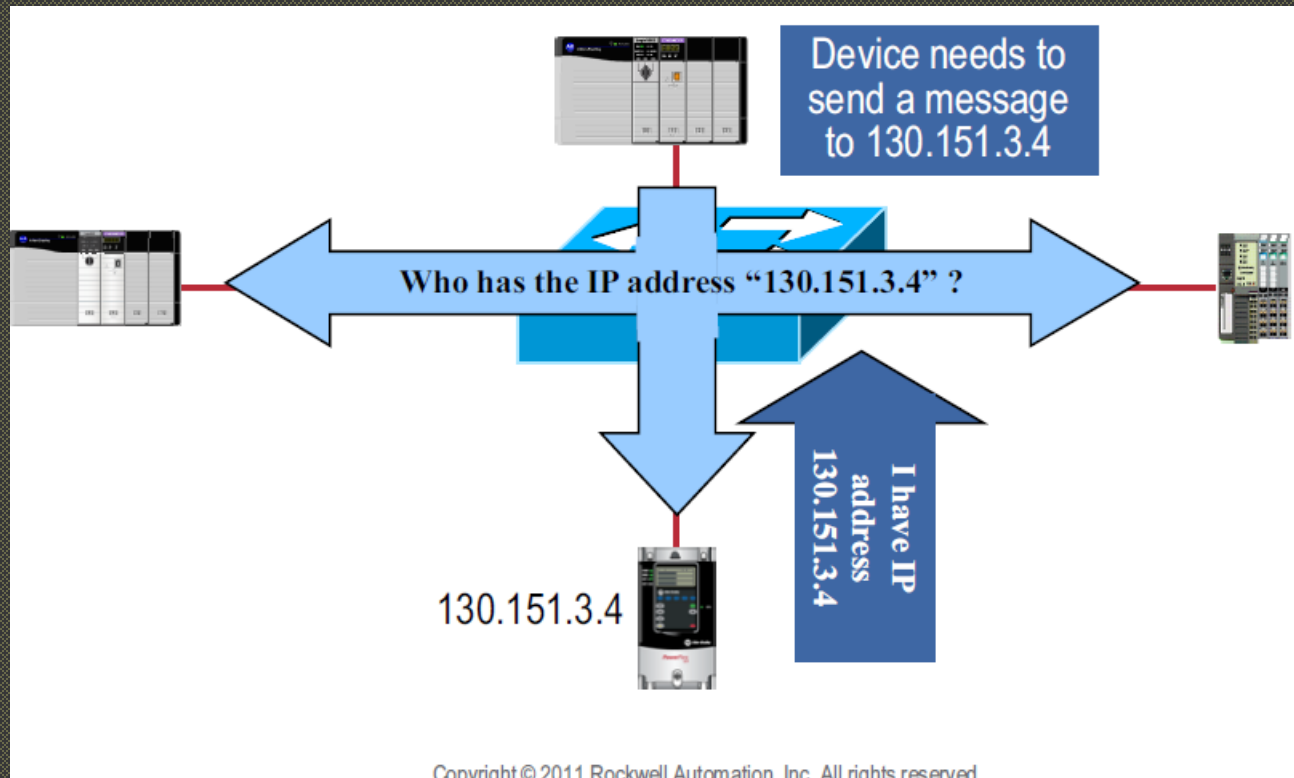
However, no single method is mandatory, leaving vendors free to implement their preferred approach. For consistency, users benefit when vendors adopt common, interoperable IP configuration mechanisms, as recommended by ODVA guidelines.



- EtherNet/IP may encounter duplicate IP address conflicts, especially when addresses are manually assigned or set to default values (e.g., 192.168.1.1).
- To handle this, EtherNet/IP can use an Address Conflict Detection (ACD) mechanism - network protocol that can be implemented in both firmware/hardware and software.

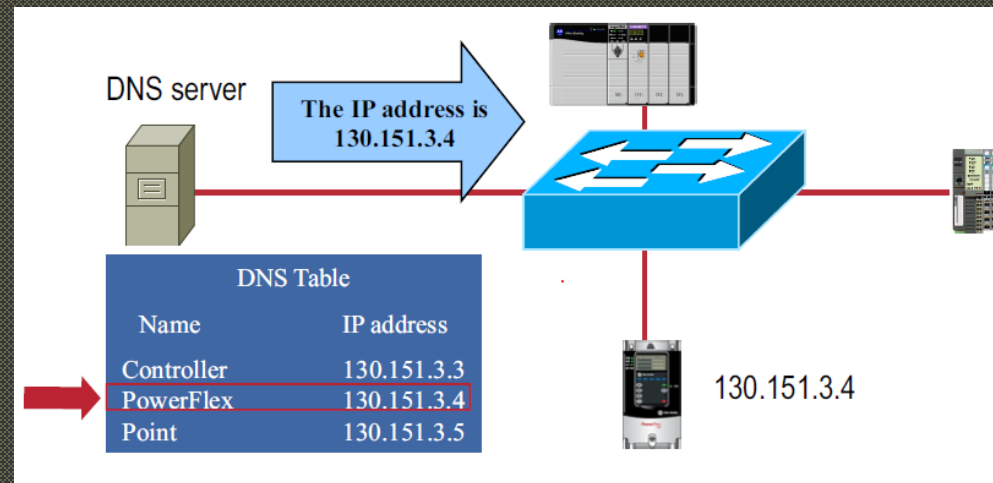
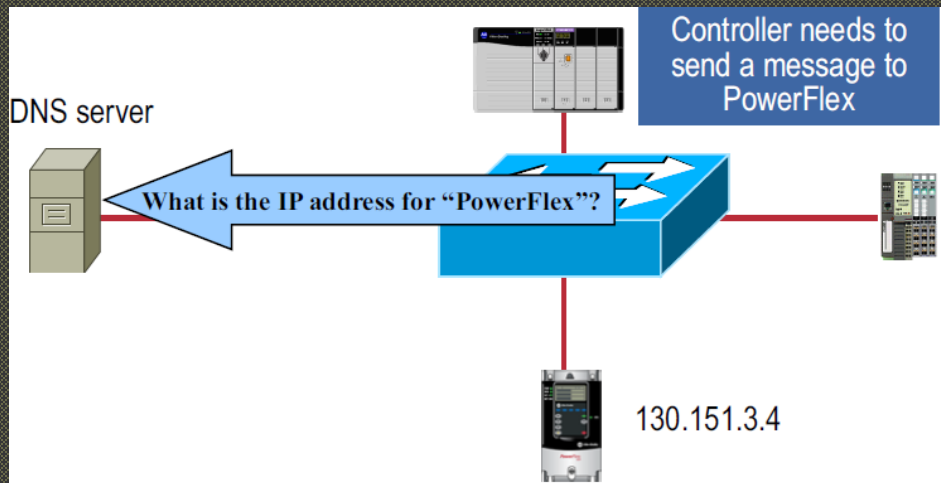
7B.12 Rockwell Software

- An ARP (Address Resolution Protocol) request is a broadcast that asks, "Who has this IP address?"
- The device with that IP replies, and the sender stores the IP-to-MAC address pair in its ARP cache.
- This allows the sender to send data directly to the correct device.



7B.12 Rockwell Software

- DNS (Domain Name System) is a protocol that lets users identify devices by names instead of IP addresses.
- A DNS server stores a table of device names and their corresponding IP addresses.
- When a device needs to contact another by name, it asks the DNS server to resolve the name to an IP address.
- The DNS server looks up its table and returns the IP address for the requested name.
- The client stores this IP address locally to avoid repeated DNS requests.
- However, the device may still need to perform an ARP request to find the hardware (MAC) address linked to that IP.



7B.12 Rockwell Software

AB_ETHIP-1\10.17.10.56-1756-EN2TR/A Configuration

General Port Configuration Advanced Port Configuration Network

Network Configuration Type

☒ Static ☐ Dynamic

☐ Use DHCP to obtain network configuration
☒ Use BOOTP to obtain network configuration

IP Address: 10 . 17 . 10 . 56

Network Mask: 255 . 255 . 255 . 0

Gateway Address: 10 . 17 . 10 . 1

Primary Name Server: 0 . 0 . 0 . 0

Secondary Name Server: 0 . 0 . 0 . 0

Domain Name:

Host Name:

Status: Network Interface Configured

OK Cancel Apply Help

RSLogix Classic
Module Configuration

RSLogix 5000
EN2TR Properties

EN2TR Webpage
Network Settings

Network Interface

Ethernet Address (MAC) 00:00:BC:2E:69:E6

IP Address 10.17.10.56

Subnet Mask 255.255.255.0

Default Gateway 10.17.10.1

Primary Name Server

Secondary Name Server

Default Domain Name

Host Name

Name Resolution DNS Enabled

SMTP Server

General Connection Module Info Time Sync RSNetWorx Internet Protocol Port Configuration Network

Internet Protocol (IP) Settings

IP settings can be manually configured or can be automatically configured if the network supports this capability.

☒ Manually configure IP settings
☐ Obtain IP settings automatically using BOOTP
☐ Obtain IP settings automatically using DHCP
☐ IP settings set by switches on the module

IP Settings Configuration

Physical Module IP Address: 10 . 17 . 10 . 56 Subnet Mask: 255 . 255 . 255 . 0

Gateway Address: 10 . 17 . 10 . 1

Domain Name:

Host Name:

Primary DNS Server Address: 0 . 0 . 0 . 0

Secondary DNS Server Address: 0 . 0 . 0 . 0

Interface Counters

Octets Inbound: 0

Octets Outbound: 0

Unicast Packets Inbound: 0

Unicast Packets Outbound: 0

Non-unicast Packets Inbound: 0

Non-unicast Packets Outbound: 0

Packets Discarded Inbound: 0

Packets Discarded Outbound: 0

Packets With Errors Inbound: 0

Packets With Errors Outbound: 0

Unknown Protocol Packets Inbound: 0

RSLogix 5000
EN2TR Properties
Port Diagnostics

EN2TR Webpage
IP Statistics

EN2TR Webpage
ARP Table

ARP Table

Physical Address	Net Address
00:00:00:00:00:00	10.17.10.50
00:00:00:00:00:00	10.17.10.51
00:00:00:00:00:00	10.17.10.52
00:00:00:00:00:00	10.17.10.54
00:00:00:00:00:00	10.17.10.57
00:00:00:00:00:00	10.17.10.58
00:00:00:00:00:00	10.17.10.60
00:00:00:00:00:00	10.17.10.62
00:00:00:00:00:00	10.17.10.64
00:00:00:00:00:00	10.17.10.66
00:00:00:00:00:00	10.17.10.70
00:00:00:00:00:00	10.17.10.72
00:24:E8:B3:8A:81	10.17.10.100
00:00:00:00:00:00	10.17.10.104

IP Statistics

Forwarding	1
Default TTL	64
In receives	89092
In header errors	0
In address errors	280
Forwarded datagrams	0
Unknown protocols	0
In discards	0
In delivers	97553
Out requests	56225
Out discards	0
Out no route	0
Reassembly timeout	20
Reassemblies requested	0
Reassemblies OK	0
Reassemblies failed	0
Reassemblies fragment OKs	0
Reassemblies fragment fails	0
Reassemblies fragment creates	0
Reassemblies routing discards	0

1756-EN2TR, an Allen-Bradley ControlLogix Ethernet/IP communication module used in industrial automation

7B.12 Rockwell Software

TCP Connections (EtherNet/IP Port)

Active	12
Maximum Observed	16
Maximum Supported	128

**EN2TR Webpage
Diagnostic Overview**

**EN2TR Webpage
UDP Statistics**

UDP Statistics

In datagrams	3
No ports	1
In errors	0
Out datagrams	4

TCP Connection Table

State	Local Address	Local Port	Remote Address	Remote Port
TIME_WAIT	10.17.10.56	80	10.17.10.100	2807
TIME_WAIT	10.17.10.56	80	10.17.10.100	2808
TIME_WAIT	10.17.10.56	80	10.17.10.100	2809
TIME_WAIT	10.17.10.56	80	10.17.10.100	2810
TIME_WAIT	10.17.10.56	80	10.17.10.100	2811
TIME_WAIT	10.17.10.56	80	10.17.10.100	2812
TIME_WAIT	10.17.10.56	80	10.17.10.100	2813
TIME_WAIT	10.17.10.56	80	10.17.10.100	2814
TIME_WAIT	10.17.10.56	80	10.17.10.100	2815
ESTABLISHED	10.17.10.56	80	10.17.10.100	2816
ESTABLISHED	10.17.10.56	44010	10.17.10.100	2004
SYN_SENT	10.17.10.56	53612	10.17.10.72	44010
SYN_SENT	10.17.10.56	53613	10.17.10.104	44010
SYN_SENT	10.17.10.56	53614	10.17.10.60	44010
SYN_SENT	10.17.10.56	53615	10.17.10.60	44010
SYN_SENT	10.17.10.56	53616	10.17.10.62	44010
SYN_SENT	10.17.10.56	53617	10.17.10.50	44010
SYN_SENT	10.17.10.56	53618	10.17.10.54	44010
SYN_SENT	10.17.10.56	53619	10.17.10.50	44010
SYN_SENT	10.17.10.56	53620	10.17.10.66	44010
SYN_SENT	10.17.10.56	53621	10.17.10.64	44010
SYN_SENT	10.17.10.56	53622	10.17.10.70	44010

**EN2TR Webpage
TCP Connection**

7B.13 Upper Layers

Layer 7 - Application Common Industrial Protocol



CIP is object-oriented, meaning each device exposes a structured set of:

- Attributes – Data values (e.g., current speed, temperature)
- Services – Functions or commands (e.g., start, stop, reset)
- Behaviors – How the object reacts to events (e.g., fault detected → shutdown)

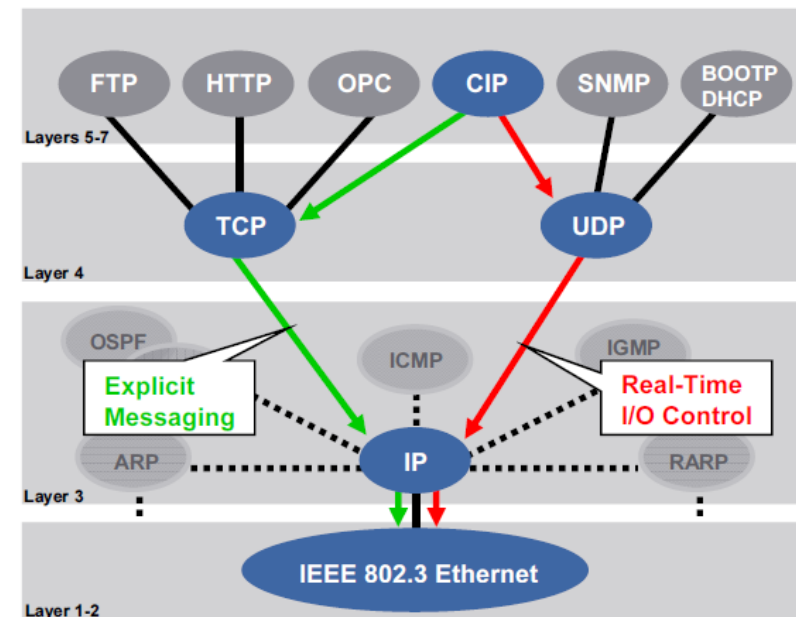
Example:

A Variable Frequency Drive (VFD) may expose a “Speed Control Object”:

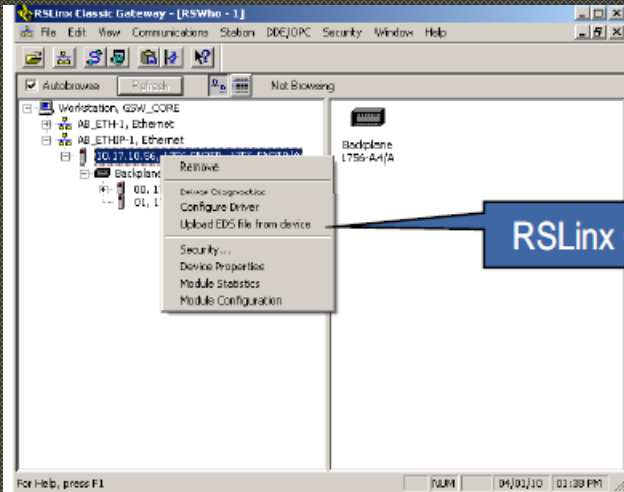
- Attribute = target RPM
- Service = set RPM
- Behavior = stop motor if overcurrent

- CIP: Implicit traffic
 - I/O control, drive control, Produced/Consumed tags
 - Uses UDP protocol (unicast and multicast)
- CIP: Explicit traffic
 - HMI, Message Instructions, Program upload/download
 - Uses TCP protocol
- Other common traffic
 - HTTP, Email, SNMP, etc.
- Advantages of EtherNet/IP
 - Standard Ethernet and IP Protocol suite
 - Future proof
 - Established – 280+ registered vendors
 - Supported – All EIP products require conformance testing

Ethernet/Industrial Protocol or EtherNet/IP specifies how CIP communication packets can be transported over standard Ethernet and TCP/IP technology.



7B.13 Upper Layers



RSLink Classic - EDS

Connection Manager

	Requests	Failed Requests	Resource Rejects	Other Rejects	Connection Timeouts
Open	0	0	0	0	0
Close	0	0	0	0	0

RSLink Classic
EN2TR Diagnostics
Connection Manager

EN2TR Webpage
Diagnostic Overview

CIP Connection Statistics (All Ports)

Active Total	0
Active Messaging	0
Active I/O	0
Maximum Total Observed	5
Maximum Total Supported	259

HMI/MSG (EtherNet/IP Port - Class 3)

Sent Packets Per Second	0
Received Packets Per Second	0
Sent Bytes Per Second	0
Received Bytes Per Second	0
Sent Packet Count	17297
Received Packet Count	17297

I/O and Prod/Cons Packets Per Second (EtherNet/IP Port - Class 1)

Total	0
Sent	0
Received	0

I/O and Prod/Cons Packet Counts (EtherNet/IP Port - Class 1)

Total	0
Sent	0
Received	0
Rejected	0
Missed	0

Multicast Producers (EtherNet/IP Port - Class 1)

Active	0
Maximum Observed	0
Maximum Supported	32
Base Address	239.192.71.224

EN2TR Webpage
Diagnostic Overview

7B.14 Modbus TCP vs EtherNet/IP

Data Representation Example

This example shows how a PLC views data from a Variable Frequency Drive (VFD). Notice how Modbus provides raw numbers requiring a manual, while EtherNet/IP provides meaningful, structured information.

Modbus TCP View

The PLC reads a list of registers. To understand this data, the programmer must consult the VFD's "Modbus Map" manual to know what each address represents and how to scale the raw values.

Address	Raw Value	Interpretation (from Manual)
40101	2	Drive Status (2 = Running)
40102	1500	Speed Reference (Value / 100 = 15.00 Hz)
40103	1498	Actual Speed (Value / 100 = 14.98 Hz)
40104	485	Motor Current (Value / 10 = 48.5 A)
40105	12	Active Fault Code (12 = Overcurrent)

EtherNet/IP View

The PLC browses the VFD's object model. The data is organized, named, and typed. The meaning is inherent to the data itself, often described electronically by an EDS file, eliminating the need for a manual.

Object/Attribute	Value
Identity.ProductName	"PowerDrive 5000"
Motor.Status	"Running" (ENUM)
Motor.SpeedRef	15.00 (REAL, Hz)
Motor.SpeedActual	14.98 (REAL, Hz)
Motor.Current	48.5 (REAL, Amps)
Fault.Code	"Overcurrent" (STRING)

55

7B.15 Electronic Data Sheet

An EDS file is like a “translator” or “instruction manual” that tells the network how to understand and talk to a device (such as a drive, sensor, or I/O module).

It follows the CIP (Common Industrial Protocol) language used in EtherNet/IP, DeviceNet, and ControlNet systems.

When you connect a new device (for example, a VFD) to an EtherNet/IP network, the controller or software needs to know:

- What the device is
- What data it can send or receive
- How to configure it

The EDS file provides all of that information in a simple, text-based format.

You can think of it like a plug-and-play ID card for the device.

Section	What It Contains	Example
[File]	Info about the EDS file itself	Version, date, author
[Device]	Device identity info	Vendor ID, Product Name, Revision
[Params] / [Groups] / [Assembly]	Configuration parameters and data organization	Motor speed, torque, etc.
[Connection Manager], [Port], [Capacity]	Network and communication setup	Number of connections, bandwidth, ports
(Optional sections)	Extra info for modular devices, languages, or specific protocols	Used in DeviceNet, EtherNet/IP, etc.

7B.15 Electronic Data Sheet

```
[Device]
DeviceType = 0x71      ; Discrete I/O Module
Vendor = 0x0001        ; Rockwell Automation
ProductName = "Discrete Input Module"
Revision = 1.0

[Object 1]              ; Identity Object
Class = 0x01
Instance = 1

[Attribute 1]
AttributeID = 0x01
AttributeName = "Vendor ID"
DataType = WORD
Access = RO

[Attribute 2]
AttributeID = 0x02
AttributeName = "Product Name"
DataType = STRING
Access = RO

[Object 2]              ; Digital Input Object
Class = 0x04
Instance = 1

[Attribute 1]
AttributeID = 0x01
AttributeName = "Input Data"
DataType = BITSTRING
Access = RO
```

[Device] Section	Meaning
DeviceType = 0x71	Tells the system what kind of device it is. Here, it's a Discrete I/O Module (a module with digital inputs or outputs).
Vendor = 0x0001	The manufacturer's ID number. 0x0001 represents Rockwell Automation.
ProductName = "Discrete Input Module"	The product's name as it will appear in the software.
Revision = 1.0	The firmware or version number of the device.

This section is like the “About” page of the device - it tells who made it, what it is, and which version it is.

[Object 1]	Meaning
Class = 0x01	Every object type has a class ID. 0x01 means this is the Identity Object, which identifies the device.
Instance = 1	The instance number. You can think of this as the specific copy of that object.

This section says, “This is the part of the device that holds its identity information.”

[Attribute 1] Section	Meaning
AttributeID = 0x01	A number that identifies this attribute.
AttributeName = "Vendor ID"	The name of the attribute, which tells us it is the vendor number.
DataType = WORD	The kind of data stored here (a WORD = 16-bit number).
Access = RO	“Read Only” - means you can view it, but cannot change it.

This tells the controller, “You can read the vendor ID from this object, but you can't modify it.”

[Attribute 2] – Product Name

- The product name is stored as a text string (for example, "Discrete Input Module").
- It is also Read Only, meaning it is just for information.

REFERENCES

The Everyman's Guide to EtherNet/IP – John S. Rinaldi

The Common Industrial Protocol (CIP™) and the Family of CIP Networks - PUB00123R1

INDUSTRIAL NETWORKING

MENG 4010

Thank you!

Discussions?